

Supplementary material for *couplr: Optimal Matching with Verifiable Certificates and Sparse Edge Generation*

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1 What this file holds

The article reports summaries of four measurements that are too large to print in full: the solver grid over problem regimes, the edge-generation loop over covariate counts, point clouds, calipers and sizes, the peak memory of every mode and comparator, and the raw runs behind the scaling table. This file carries them in full, and names the script that produced each one so any of them can be re-measured. It also carries the parts of the software design the article points at: the object model, the C++ organisation and the dependency footprint, the proof of the sentinel-padding lemma the article states, and the outcome analysis the article points at.

```
## run          20260830-152628
## host         Darwin Mac 25.5.0 Darwin Kernel Version 25.5.0: Mon Apr 27 20:41:15 PDT 2026; root:xnu-12
## commit       a63b760 28 modified paths
## R version    4.5.3 (2026-03-11)
## aarch64-apple-darwin25.3.0
## couplr       1.7.0
```

```

## MatchIt          4.7.2
## optmatch         0.10.8
## microbenchmark  1.5.0
##
## note             Every arm the article reports was measured at this one run and
##                   commit. The harness read 1.6.2 from the DESCRIPTION at a63b760 and
##                   the version is recorded here as 1.7.0: no commit between a63b760 and
##                   the 1.7.0 release touches a file under R/ or src/, so the sources
##                   that produced these numbers are the released ones and only the
##                   version string differs. The modified paths the run recorded are the
##                   benchmark outputs it was writing, together with two RcppExports
##                   files that differ from the commit only in line endings.

```

Section	Script	Files
Solver selection across regimes	scripts/bench_regimes.R	regime-runs.csv, regime-cells.csv, regime-results.csv, regime-verdict.csv
The edge-generation loop	scripts/bench_implicit_grid.R	implicit-grid-runs.csv, implicit-grid-results.csv
Peak memory	scripts/bench_memory.R	memory-results.csv
Representation against solver	scripts/bench_implicit.R	implicit-results.csv
Scaling, every run	scripts/bench_scaling.R	scaling-runs.csv, scaling-results.csv

Every script is resume-safe: it writes its rows as it goes and skips a cell it already holds, so a run that is interrupted continues where it stopped. Moving the CSV aside forces a full re-measurement. `paper/run_bench_suite.sh` runs all of them in order on one machine, which is how the numbers here were produced.

The scripts load `couplr` from a source checkout with `pkgload::load_all()` and write their CSVs under `paper/`, so they run from a clone of the package repository rather than from the submission archive:

```

git clone https://github.com/gcol33/couplr
cd couplr
sh paper/run_bench_suite.sh

```

The scaling, implicit, implicit-grid, regimes and memory scripts also take `--quick`, which runs a reduced grid.

2 The sentinel padding lemma

Tight constraints can leave the matching problem infeasible, and `couplr` answers it by replacing the forbidden entries with a finite sentinel and re-solving. The article states the lemma that makes a minimum-cost solve of the padded problem return the lexicographic optimum, the largest admissible matching and the cheapest among those of its size, and defers the proof here.

Write $[\ell, h]$ for the range of the admissible costs of the pruned submatrix, k for the smaller of its two dimensions and σ for the sentinel. Every feasible assignment of the padded problem matches all k rows, so it splits into s sentinel edges and $k - s$ real ones and costs $s\sigma + R$ with $R \in [(k - s)\ell, (k - s)h]$.

Lemma 1. *If $\sigma > k(|\ell| + |h|)$, then a padded assignment using more sentinel edges than another costs strictly more.*

Proof. Let M' and M use s' and s sentinel edges with $s' > s$, and write R' and R for their real costs. Then

$$\text{cost}(M') - \text{cost}(M) = (s' - s)\sigma + R' - R \geq \sigma + (k - s')\ell - (k - s)h \geq \sigma - k|\ell| - k|h| > 0. \quad \square$$

couplr takes $\sigma = (k + 1)(|\ell| + |h|) + 1$, which satisfies the hypothesis whatever the signs of ℓ and h and when the admissible costs are all zero, and leaves a margin of $|\ell| + |h| + 1$ between a matching and one using more sentinel edges. `assignment(cardinality = "maximum")` prices its dummy columns with the same quantity, and a maximisation is solved by negation with the sentinel negated alongside it, so the lemma applies to the negated instance. The reported objective is recomputed over the real pairs, so no sentinel enters it.

The lemma's hypothesis is a statement about the numbers actually used, so it is checkable on the doubles in hand and couplr's σ satisfies it by construction. What the lemma does not settle is the solve. A solver working where a double no longer resolves a gap of $|\ell| + |h| + 1$ cannot be relied on to order two assignments the padding separates by that much, so couplr refuses a padded objective above 2^{53} rather than returning an ordering its arithmetic may not support. That threshold is a precondition on the solve and not a proof of it: 2^{53} bounds the integers a double carries exactly, and the padded costs need not be integers. What establishes that a particular padded solve reached its optimum is the certificate `verify_assignment()` returns for the padded matrix, and the lemma carries that verdict across to the lexicographic objective.

3 From matched pairs to an estimate

The article shows the matched sample and defers the outcome analysis here. The match is rebuilt below so this file stands on its own.

```
library(couplr)
data(hospital_staff)
treated <- transform(hospital_staff$nurses_extended, id = nurse_id)
control <- transform(hospital_staff$controls_extended, id = nurse_id)
covars <- c("age", "experience_years", "certification_level")
m <- match_couples(treated, control, vars = covars, auto_scale = TRUE)
```

Taking the hourly rate as the outcome, the paired difference within each matched pair estimates the association between full-time status and pay among nurses comparable on age, experience and certification.

```
matched <- join_matched(m, treated, control)
d <- matched$hourly_rate_left - matched$hourly_rate_right
tt <- t.test(d)
c(estimate = mean(d), lower = tt$conf.int[1], upper = tt$conf.int[2])
```

```
## estimate    lower    upper
## 3.531350 1.875651 5.187049
```

Full-time nurses earn 3.53 more per hour than their matched part-time counterparts, with a 95 percent interval that excludes zero. As a paired difference on a matched sample this estimates an adjusted association. Reading it as the causal effect of full-time status is what rests on assumptions no matching method can verify: conditional ignorability, positivity, and the stable unit treatment value assumption (SUTVA), under which a unit's outcome depends on its own treatment alone. The matched output can also be passed to an outcome model for regression adjustment; valid estimation and uncertainty quantification still depend on the estimator used.

What matching can offer against the first of those assumptions is a statement about how strong an unmeasured confounder would have to be to overturn the result. `sensitivity_analysis()` implements Rosenbaum bounds (Rosenbaum 2002) for the Wilcoxon signed-rank statistic on one-to-one matched pairs, reporting upper and lower p-value bounds across a grid of the odds-ratio parameter Γ .

```
sens <- sensitivity_analysis(m, treated, control, outcome_var = "hourly_rate")
sens$results[sens$results$gamma %in% c(1, 1.5, 1.75, 2), ]
```

```
## # A tibble: 4 x 4
##   gamma t_stat p_upper p_lower
##   <dbl> <dbl>   <dbl>   <dbl>
```

```
## 1 1 13304. 0.0000187 1.87e- 5
## 2 1.5 13304. 0.0435 1.01e-11
## 3 1.75 13304. 0.207 4.48e-15
## 4 2 13304. 0.481 1.69e-18
```

At $\Gamma = 1$, meaning no hidden bias, the association is significant. It survives an unmeasured confounder that raises one pair member's odds of being full-time by half, and stops being significant at $\Gamma = 1.75$. A reader can judge from that number whether a confounder of that strength is plausible in this setting; the analysis quantifies exposure to hidden bias rather than removing it.

4 Solver selection across problem regimes

`assignment(method = "auto")` chooses a solver from five rules read off one pass over the cost matrix: enumerate an at most 8 by 8 problem, use a cardinality algorithm where the finite costs are constant or binary, use the sparse Jonker-Volgenant variant where more than half the entries are forbidden, use the shortest-augmenting-path solver where there are at least three times as many columns as rows, and otherwise use Jonker-Volgenant. The article's solver figure varies problem size on square dense integer costs, which is the last rule's regime and no other. This grid varies the properties the rules are stated over.

A cell is a cost regime, a forbidden pattern and a shape. Each cell is generated as several independent instances, each instance is timed several times, and every solver in the panel sees the same instance.

Table 1: The factors the solver grid crosses. Shapes and sizes are in the next table.

Factor	Level	Meaning
cost regime	int_uniform	integer, uniform on 1..10,000
cost regime	dbl_uniform	double, uniform on (0, 1)
cost regime	binary	binary, 0 or 1
cost regime	constant	constant
cost regime	tied	integer, five distinct values
cost regime	heavy_tailed	double, lognormal
cost regime	metric_uniform	euclidean over uniform points in 8 dimensions
cost regime	metric_clustered	euclidean over eight clusters in 8 dimensions
forbidden pattern	none	complete: every pair admissible
forbidden pattern	random_60	60 percent of pairs admissible, unstructured
forbidden pattern	random_25	25 percent admissible, unstructured
forbidden pattern	block_4	25 percent admissible in four disconnected components

Table 2: Shapes. The base tier runs the whole panel at a size every solver in it reaches; the large tier repeats the crossing at a size only the scalable solvers reach.

tier	Shape	aspect	Solvers in the panel
base	500x500	1:1	12
base	500x1500	1:3	12
base	500x5000	1:10	12
large	1000x10000	1:10	5
large	1500x1500	1:1	5
large	1500x4500	1:3	5

4.1 What the rules cost

For each cell, the solver "auto" selected is timed as a solver of its own, so its number carries the probing pass the rules cost, and it is compared against the fastest named solver in the same cell.

Table 3: Each dispatch rule, over the cells where it fired. Ratio is the time the dispatched solver took over the time the cell's fastest named solver took, so 1.00x means the rule picked the best available solver and 2.00x means it cost twice the best.

Rule	Cells	Picked the cell's fastest	Median ratio	Worst ratio
default	144	59	1.04x	9.95x
no_cost_scale	45	35	1.00x	1.93x

Table 4: The 25 cells where the dispatched solver was furthest off the cell's fastest.

Regime	Pattern	Shape	Rule	Picked	Best	Dispatched (ms)	Best (ms)	Ratio
tied	25%	500x500	default	jv	auction_scaled	63.6	6.4	9.95x
tied	random_05500x500	500x500	default	jv	auction_scaled	41.5	4.2	9.91x
tied	complete	500x500	default	jv	network_simplex	57.9	6.5	8.87x
tied	complete	500x1500	default	jv	network_simplex	204	23.4	8.72x
tied	60%	500x500	default	jv	network_simplex	63.9	8.2	7.82x
tied	25%	500x1500	default	jv	network_simplex	203	37.2	5.45x
tied	60%	500x1500	default	jv	network_simplex	209	38.6	5.41x
tied	complete	500x5000	default	jv	network_simplex	700	132	5.29x
tied	4 blocks	500x500	default	jv	auction_scaled	17.9	4.1	4.32x
int_uniform	random_01500x500	500x500	default	jv	auction_scaled	5.5	1.4	3.89x
dbl_uniform	random_01500x500	500x500	default	jv	auction_scaled	6.4	1.7	3.67x
tied	60%	500x5000	default	jv	network_simplex	716	197	3.64x
metric_clustered	random_05500x500	500x500	default	jv	auction_scaled	9	2.5	3.56x
metric_uniform	random_05500x500	500x500	default	jv	auction_scaled	8.1	2.3	3.51x
metric_clustered	random_01500x500	500x500	default	jv	csa	5.4	1.6	3.46x
tied	random_05500x5000	500x500	default	jv	csflow	386	113	3.43x
heavy_tailed	random_01500x500	500x500	default	jv	auction_scaled	7.6	2.3	3.23x
heavy_tailed	random_011500x1500	500x1500	default	jv	sap	112	35.2	3.18x
dbl_uniform	random_05500x500	500x500	default	jv	auction_scaled	7.4	2.3	3.17x
int_uniform	random_05500x500	500x500	default	jv	auction_scaled	7.2	2.3	3.06x
metric_uniform	random_01500x500	500x500	default	jv	auction_scaled	5.5	1.8	3.06x
tied	25%	500x5000	default	jv	network_simplex	704	236	2.99x
tied	random_05500x1500	500x1500	default	jv	csflow	114	39.6	2.89x
int_uniform	random_011500x1500	500x1500	default	jv	sap	89.2	30.9	2.89x
metric_clustered	25%	500x500	default	jv	auction_scaled	14	4.9	2.82x

4.2 Every cell

Table 5: Every cell of the solver grid: the rule that fired, the solver it picked, the fastest solver in the cell, and the ratio between them.

Regime	Pattern	Shape	Rule	Picked	Best	Ratio
binary	4 blocks	500x1500	no_cost_scale	hk01	hk01	1.03x
binary	4 blocks	500x500	no_cost_scale	hk01	hk01	1.02x
binary	4 blocks	500x5000	no_cost_scale	hk01	hk01	1.20x
binary	complete	500x1500	no_cost_scale	hk01	hk01	1.02x
binary	complete	500x500	no_cost_scale	hk01	hk01	1.00x
binary	complete	500x5000	no_cost_scale	hk01	hk01	1.00x
binary	random_01	500x1500	no_cost_scale	hk01	hk01	0.72x
binary	random_01	500x500	no_cost_scale	hk01	auction_scaled	1.93x
binary	random_01	500x5000	no_cost_scale	hk01	hk01	0.75x
binary	random_05	500x1500	no_cost_scale	hk01	hk01	0.99x
binary	random_05	500x500	no_cost_scale	hk01	hk01	1.00x
binary	random_05	500x5000	no_cost_scale	hk01	hk01	1.33x
binary	25%	500x1500	no_cost_scale	hk01	hk01	1.03x
binary	25%	500x500	no_cost_scale	hk01	hk01	1.02x
binary	25%	500x5000	no_cost_scale	hk01	hk01	1.14x
binary	60%	500x1500	no_cost_scale	hk01	hk01	0.97x
binary	60%	500x500	no_cost_scale	hk01	hk01	1.00x
binary	60%	500x5000	no_cost_scale	hk01	hk01	0.95x
constant	4 blocks	500x1500	no_cost_scale	hk01	hk01	1.00x
constant	4 blocks	500x500	no_cost_scale	hk01	hk01	1.00x
constant	4 blocks	500x5000	no_cost_scale	hk01	hk01	1.00x
constant	complete	500x1500	no_cost_scale	hk01	hk01	1.00x
constant	complete	500x500	no_cost_scale	hk01	hk01	1.01x
constant	complete	500x5000	no_cost_scale	hk01	hk01	1.00x
constant	random_01	500x1500	no_cost_scale	hk01	hk01	1.38x
constant	random_01	500x500	no_cost_scale	hk01	hk01	1.38x
constant	random_01	500x5000	no_cost_scale	hk01	hk01	1.33x
constant	random_05	500x1500	no_cost_scale	hk01	hk01	1.35x
constant	random_05	500x500	no_cost_scale	hk01	hk01	1.01x
constant	random_05	500x5000	no_cost_scale	hk01	hk01	1.35x
constant	25%	500x1500	no_cost_scale	hk01	hk01	1.17x
constant	25%	500x500	no_cost_scale	hk01	hk01	1.18x
constant	25%	500x5000	no_cost_scale	hk01	hk01	1.00x
constant	60%	500x1500	no_cost_scale	hk01	hk01	1.07x
constant	60%	500x500	no_cost_scale	hk01	hk01	1.03x
constant	60%	500x5000	no_cost_scale	hk01	hk01	1.01x
dbl_uniform	4 blocks	500x1500	default	jv	hungarian	1.26x
dbl_uniform	4 blocks	500x500	default	jv	auction_scaled	1.39x
dbl_uniform	4 blocks	500x5000	default	jv	jv	0.99x
dbl_uniform	complete	500x1500	default	jv	hungarian	1.04x
dbl_uniform	complete	500x500	default	jv	jv	1.00x
dbl_uniform	complete	500x5000	default	jv	hungarian	1.00x
dbl_uniform	random_01	500x1500	default	jv	hungarian	1.34x
dbl_uniform	random_01	500x500	default	jv	auction_scaled	3.67x
dbl_uniform	random_01	500x5000	default	jv	csflow	0.94x
dbl_uniform	random_05	500x1500	default	jv	lapmod	1.17x
dbl_uniform	random_05	500x500	default	jv	auction_scaled	3.17x
dbl_uniform	random_05	500x5000	default	jv	csflow	1.02x
dbl_uniform	25%	500x1500	default	jv	jv	0.99x

Regime	Pattern	Shape	Rule	Picked	Best	Ratio
dbl_uniform	25%	500x500	default	jv	auction_scaled	1.96x
dbl_uniform	25%	500x5000	default	jv	jv	1.19x
dbl_uniform	60%	500x1500	default	jv	jv	1.01x
dbl_uniform	60%	500x500	default	jv	auction_scaled	1.20x
dbl_uniform	60%	500x5000	default	jv	hungarian	0.93x
heavy_tailed	4 blocks	500x1500	default	jv	jv	1.01x
heavy_tailed	4 blocks	500x500	default	jv	hungarian	1.05x
heavy_tailed	4 blocks	500x5000	default	jv	hungarian	1.02x
heavy_tailed	complete	500x1500	default	jv	jv	1.00x
heavy_tailed	complete	500x500	default	jv	jv	1.00x
heavy_tailed	complete	500x5000	default	jv	hungarian	1.01x
heavy_tailed	random_01	500x1500	default	jv	hungarian	1.02x
heavy_tailed	random_01	500x500	default	jv	auction_scaled	3.23x
heavy_tailed	random_01	500x5000	default	jv	hungarian	1.34x
heavy_tailed	random_05	500x1500	default	jv	lapmod	1.18x
heavy_tailed	random_05	500x500	default	jv	auction_scaled	2.31x
heavy_tailed	random_05	500x5000	default	jv	hungarian	1.27x
heavy_tailed	25%	500x1500	default	jv	hungarian	0.99x
heavy_tailed	25%	500x500	default	jv	auction_scaled	1.29x
heavy_tailed	25%	500x5000	default	jv	hungarian	1.20x
heavy_tailed	60%	500x1500	default	jv	hungarian	1.01x
heavy_tailed	60%	500x500	default	jv	jv	1.00x
heavy_tailed	60%	500x5000	default	jv	jv	1.02x
int_uniform	4 blocks	500x1500	default	jv	hungarian	1.01x
int_uniform	4 blocks	500x500	default	jv	csa	1.20x
int_uniform	4 blocks	500x5000	default	jv	jv	1.27x
int_uniform	complete	500x1500	default	jv	jv	1.02x
int_uniform	complete	500x500	default	jv	jv	1.00x
int_uniform	complete	500x5000	default	jv	hungarian	1.02x
int_uniform	random_01	500x1500	default	jv	jv	1.02x
int_uniform	random_01	500x500	default	jv	auction_scaled	3.89x
int_uniform	random_01	500x5000	default	jv	hungarian	1.01x
int_uniform	random_05	500x1500	default	jv	csflow	1.02x
int_uniform	random_05	500x500	default	jv	auction_scaled	3.06x
int_uniform	random_05	500x5000	default	jv	hungarian	1.01x
int_uniform	25%	500x1500	default	jv	jv	1.00x
int_uniform	25%	500x500	default	jv	auction_scaled	1.65x
int_uniform	25%	500x5000	default	jv	hungarian	1.04x
int_uniform	60%	500x1500	default	jv	hungarian	1.01x
int_uniform	60%	500x500	default	jv	auction_scaled	1.11x
int_uniform	60%	500x5000	default	jv	hungarian	1.07x
metric_clustered	4 blocks	500x1500	default	jv	jv	0.99x
metric_clustered	4 blocks	500x500	default	jv	auction_scaled	1.80x
metric_clustered	4 blocks	500x5000	default	jv	hungarian	1.07x
metric_clustered	complete	500x1500	default	jv	jv	1.00x
metric_clustered	complete	500x500	default	jv	jv	0.90x
metric_clustered	complete	500x5000	default	jv	hungarian	1.01x
metric_clustered	random_01	500x1500	default	jv	hungarian	1.00x
metric_clustered	random_01	500x500	default	jv	csa	3.46x
metric_clustered	random_01	500x5000	default	jv	csflow	1.22x
metric_clustered	random_05	500x1500	default	jv	jv	1.01x
metric_clustered	random_05	500x500	default	jv	auction_scaled	3.56x

Regime	Pattern	Shape	Rule	Picked	Best	Ratio
metric_clustered	random_05	500x5000	default	jv	csflow	1.03x
metric_clustered	25%	500x1500	default	jv	jv	1.15x
metric_clustered	25%	500x500	default	jv	auction_scaled	2.82x
metric_clustered	25%	500x5000	default	jv	jv	1.14x
metric_clustered	60%	500x1500	default	jv	jv	1.00x
metric_clustered	60%	500x500	default	jv	auction_scaled	1.76x
metric_clustered	60%	500x5000	default	jv	jv	1.09x
metric_uniform	4 blocks	500x1500	default	jv	jv	0.99x
metric_uniform	4 blocks	500x500	default	jv	auction_scaled	1.82x
metric_uniform	4 blocks	500x5000	default	jv	hungarian	1.02x
metric_uniform	complete	500x1500	default	jv	jv	1.01x
metric_uniform	complete	500x500	default	jv	jv	0.99x
metric_uniform	complete	500x5000	default	jv	hungarian	1.01x
metric_uniform	random_01	500x1500	default	jv	hungarian	1.01x
metric_uniform	random_01	500x500	default	jv	auction_scaled	3.06x
metric_uniform	random_01	500x5000	default	jv	csflow	1.20x
metric_uniform	random_05	500x1500	default	jv	csflow	1.02x
metric_uniform	random_05	500x500	default	jv	auction_scaled	3.51x
metric_uniform	random_05	500x5000	default	jv	hungarian	0.78x
metric_uniform	25%	500x1500	default	jv	hungarian	1.17x
metric_uniform	25%	500x500	default	jv	auction_scaled	2.69x
metric_uniform	25%	500x5000	default	jv	jv	1.19x
metric_uniform	60%	500x1500	default	jv	jv	1.00x
metric_uniform	60%	500x500	default	jv	auction_scaled	1.47x
metric_uniform	60%	500x5000	default	jv	hungarian	1.01x
tied	4 blocks	500x1500	default	jv	csflow	1.78x
tied	4 blocks	500x500	default	jv	auction_scaled	4.32x
tied	4 blocks	500x5000	default	jv	csflow	2.01x
tied	complete	500x1500	default	jv	network_simplex	8.72x
tied	complete	500x500	default	jv	network_simplex	8.87x
tied	complete	500x5000	default	jv	network_simplex	5.29x
tied	random_01	500x1500	default	jv	csflow	1.08x
tied	random_01	500x500	default	jv	auction_scaled	2.54x
tied	random_01	500x5000	default	jv	jv	1.30x
tied	random_05	500x1500	default	jv	csflow	2.89x
tied	random_05	500x500	default	jv	auction_scaled	9.91x
tied	random_05	500x5000	default	jv	csflow	3.43x
tied	25%	500x1500	default	jv	network_simplex	5.45x
tied	25%	500x500	default	jv	auction_scaled	9.95x
tied	25%	500x5000	default	jv	network_simplex	2.99x
tied	60%	500x1500	default	jv	network_simplex	5.41x
tied	60%	500x500	default	jv	network_simplex	7.82x
tied	60%	500x5000	default	jv	network_simplex	3.64x
binary	complete	1000x10000	no_cost_scale	hk01	jv	0.03x
binary	complete	1500x1500	no_cost_scale	hk01	ramshaw_tarjan	0.02x
binary	complete	1500x4500	no_cost_scale	hk01	jv	0.02x
binary	random_01	1000x10000	no_cost_scale	hk01	sap	0.26x
binary	random_01	1500x1500	no_cost_scale	hk01	sap	0.08x
binary	random_01	1500x4500	no_cost_scale	hk01	sap	0.14x
binary	25%	1000x10000	no_cost_scale	hk01	sap	0.07x
binary	25%	1500x1500	no_cost_scale	hk01	sap	0.03x
binary	25%	1500x4500	no_cost_scale	hk01	sap	0.05x

Regime	Pattern	Shape	Rule	Picked	Best	Ratio
dbl_uniform	complete	1000x10000	default	jv	jv	0.99x
dbl_uniform	complete	1500x1500	default	jv	jv	1.00x
dbl_uniform	complete	1500x4500	default	jv	jv	1.01x
dbl_uniform	random_01	1000x10000	default	jv	jv	0.96x
dbl_uniform	random_01	1500x1500	default	jv	sap	2.62x
dbl_uniform	random_01	1500x4500	default	jv	jv	1.10x
dbl_uniform	25%	1000x10000	default	jv	jv	0.97x
dbl_uniform	25%	1500x1500	default	jv	jv	1.02x
dbl_uniform	25%	1500x4500	default	jv	jv	1.01x
heavy_tailed	complete	1000x10000	default	jv	jv	1.00x
heavy_tailed	complete	1500x1500	default	jv	jv	1.00x
heavy_tailed	complete	1500x4500	default	jv	jv	0.99x
heavy_tailed	random_01	1000x10000	default	jv	jv	0.97x
heavy_tailed	random_01	1500x1500	default	jv	sap	3.18x
heavy_tailed	random_01	1500x4500	default	jv	jv	1.11x
heavy_tailed	25%	1000x10000	default	jv	jv	0.96x
heavy_tailed	25%	1500x1500	default	jv	jv	1.02x
heavy_tailed	25%	1500x4500	default	jv	jv	1.01x
int_uniform	complete	1000x10000	default	jv	jv	1.00x
int_uniform	complete	1500x1500	default	jv	jv	1.00x
int_uniform	complete	1500x4500	default	jv	jv	1.01x
int_uniform	random_01	1000x10000	default	jv	jv	1.05x
int_uniform	random_01	1500x1500	default	jv	sap	2.89x
int_uniform	random_01	1500x4500	default	jv	jv	0.97x
int_uniform	25%	1000x10000	default	jv	jv	1.00x
int_uniform	25%	1500x1500	default	jv	jv	1.00x
int_uniform	25%	1500x4500	default	jv	jv	1.05x
metric_clustered	complete	1000x10000	default	jv	jv	1.01x
metric_clustered	complete	1500x1500	default	jv	jv	1.00x
metric_clustered	complete	1500x4500	default	jv	jv	1.00x
metric_clustered	random_01	1000x10000	default	jv	jv	1.04x
metric_clustered	random_01	1500x1500	default	jv	sap	2.76x
metric_clustered	random_01	1500x4500	default	jv	lapmod	1.05x
metric_clustered	25%	1000x10000	default	jv	jv	0.96x
metric_clustered	25%	1500x1500	default	jv	sap	1.05x
metric_clustered	25%	1500x4500	default	jv	jv	1.00x

4.3 Every solver, every cell

Table 6: Milliseconds per solver and cell: the median across instances, with the quartiles beside it. An instance contributes the median of its own repetitions, so the quartiles are between problems rather than between repetitions of one problem. A solver that errored or timed out on a cell has no row there.

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	4 blocks	500x500	hk01	3	2.02	1.86	2.03
binary	4 blocks	500x500	auto	3	2.06	1.87	2.06
binary	4 blocks	500x500	auction_scaled	3	4.92	4.78	4.95
binary	4 blocks	500x500	csa	3	5.16	5.14	5.58

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	4 blocks	500x500	network_simplex	3	5.48	5.19	5.7
binary	4 blocks	500x500	sap	3	11.4	11.3	11.4
binary	4 blocks	500x500	csflow	3	11.7	11.7	12.3
binary	4 blocks	500x500	ramshaw_tarjan	3	15	14.9	15.4
binary	4 blocks	500x500	jv	3	17.7	17.6	17.8
binary	4 blocks	500x500	hungarian	3	17.9	17.7	18
binary	4 blocks	500x500	lapmod	3	18.5	18.3	18.5
binary	4 blocks	500x500	sap_dense	3	26.3	26.3	26.4
binary	4 blocks	500x1500	hk01	3	4.76	4.76	4.78
binary	4 blocks	500x1500	auto	3	4.92	4.84	5.31
binary	4 blocks	500x1500	csflow	3	26.4	26.4	26.5
binary	4 blocks	500x1500	sap	3	26.7	26.6	27.1
binary	4 blocks	500x1500	network_simplex	3	36.7	34.8	37.6
binary	4 blocks	500x1500	ramshaw_tarjan	3	40.9	40.5	41
binary	4 blocks	500x1500	hungarian	3	53.6	53.5	54.1
binary	4 blocks	500x1500	jv	3	53.6	53.6	53.6
binary	4 blocks	500x1500	lapmod	3	56.7	56.6	57.1
binary	4 blocks	500x1500	sap_dense	3	78.6	78.6	79
binary	4 blocks	500x1500	csa	3	103	97.8	103
binary	4 blocks	500x1500	auction_scaled	3	160	158	163
binary	4 blocks	500x5000	hk01	3	16.1	16.1	17.7
binary	4 blocks	500x5000	auto	3	19.3	17.7	21.2
binary	4 blocks	500x5000	sap	3	78	77.9	79.6
binary	4 blocks	500x5000	csflow	3	78.2	78.1	78.4
binary	4 blocks	500x5000	ramshaw_tarjan	3	133	132	137
binary	4 blocks	500x5000	hungarian	3	183	181	184
binary	4 blocks	500x5000	jv	3	183	183	184
binary	4 blocks	500x5000	network_simplex	3	188	188	197
binary	4 blocks	500x5000	lapmod	3	192	192	193
binary	4 blocks	500x5000	sap_dense	3	261	261	261
binary	4 blocks	500x5000	csa	3	1.37e+03	1.37e+03	1.39e+03
binary	4 blocks	500x5000	auction_scaled	3	2.08e+03	2.08e+03	2.16e+03
binary	complete	500x500	auto	3	3.73	3.72	3.74
binary	complete	500x500	hk01	3	3.74	3.72	3.75
binary	complete	500x500	network_simplex	3	5	4.23	5.03
binary	complete	500x500	auction_scaled	3	14.5	14.4	14.5
binary	complete	500x500	csa	3	14.5	14.5	14.8
binary	complete	500x500	jv	3	59	58.9	59.2
binary	complete	500x500	hungarian	3	60.1	60	60.4
binary	complete	500x500	ramshaw_tarjan	3	64	64	64.1
binary	complete	500x500	sap_dense	3	90.5	90	91.2
binary	complete	500x500	lapmod	3	96	94.7	96.2
binary	complete	500x500	sap	3	107	107	107
binary	complete	500x500	csflow	3	108	108	108
binary	complete	500x1500	hk01	3	10.6	10.6	10.7
binary	complete	500x1500	auto	3	10.8	10.8	10.9
binary	complete	500x1500	network_simplex	3	25.5	24.2	30.6
binary	complete	500x1500	csa	3	88	83.3	89.1
binary	complete	500x1500	auction_scaled	3	129	128	129
binary	complete	500x1500	jv	3	199	199	200
binary	complete	500x1500	hungarian	3	201	200	202

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	complete	500x1500	ramshaw_tarjan	3	211	211	212
binary	complete	500x1500	sap	3	256	255	256
binary	complete	500x1500	csflow	3	260	259	260
binary	complete	500x1500	sap_dense	3	284	284	284
binary	complete	500x1500	lapmod	3	293	293	293
binary	complete	500x5000	hk01	3	35.5	35.5	35.5
binary	complete	500x5000	auto	3	35.6	35.5	35.6
binary	complete	500x5000	network_simplex	3	112	103	116
binary	complete	500x5000	jv	3	683	683	684
binary	complete	500x5000	hungarian	3	684	683	684
binary	complete	500x5000	ramshaw_tarjan	3	755	754	763
binary	complete	500x5000	sap	3	826	817	827
binary	complete	500x5000	csflow	3	873	872	874
binary	complete	500x5000	sap_dense	3	968	968	969
binary	complete	500x5000	lapmod	3	996	995	996
binary	complete	500x5000	csa	3	1.11e+03	1.07e+03	1.25e+03
binary	complete	500x5000	auction_scaled	3	1.69e+03	1.69e+03	1.74e+03
binary	random_01	500x500	auction_scaled	3	2.36	2.31	2.45
binary	random_01	500x500	csa	3	2.58	2.41	3.13
binary	random_01	500x500	auto	3	4.56	4.54	5.46
binary	random_01	500x500	sap	3	4.59	4.56	5.22
binary	random_01	500x500	csflow	3	4.6	4.15	5.47
binary	random_01	500x500	hk01	3	4.9	4.75	6.08
binary	random_01	500x500	jv	3	10.7	10.6	13.1
binary	random_01	500x500	lapmod	3	11	10.2	13.4
binary	random_01	500x500	hungarian	3	11.1	10.3	13.3
binary	random_01	500x500	ramshaw_tarjan	3	11.1	10.4	13.3
binary	random_01	500x500	sap_dense	3	17.1	16.5	21.1
binary	random_01	500x500	network_simplex	3	24.2	21.8	25.7
binary	random_01	500x1500	auto	3	3.31	3.3	3.94
binary	random_01	500x1500	hk01	3	4.58	3.95	4.6
binary	random_01	500x1500	sap	3	5.8	5.36	6.21
binary	random_01	500x1500	csflow	3	6.07	5.83	6.35
binary	random_01	500x1500	ramshaw_tarjan	3	11	10.9	12.4
binary	random_01	500x1500	hungarian	3	11.9	11	13
binary	random_01	500x1500	jv	3	11.9	10.4	13
binary	random_01	500x1500	lapmod	3	12.1	10.5	13
binary	random_01	500x1500	sap_dense	3	24.1	20.7	24.1
binary	random_01	500x1500	network_simplex	3	34	33.7	34.2
binary	random_01	500x1500	csa	3	128	116	139
binary	random_01	500x1500	auction_scaled	3	196	193	199
binary	random_01	500x5000	auto	3	12.1	12.1	14.1
binary	random_01	500x5000	hk01	3	16.2	14	16.5
binary	random_01	500x5000	csflow	3	19.8	18.9	21
binary	random_01	500x5000	sap	3	24.5	23.6	24.9
binary	random_01	500x5000	ramshaw_tarjan	3	44.5	44	48
binary	random_01	500x5000	lapmod	3	45.1	42.5	50
binary	random_01	500x5000	hungarian	3	45.6	45.1	48.6
binary	random_01	500x5000	jv	3	47.7	46.2	51.9
binary	random_01	500x5000	sap_dense	3	76	74	83.5
binary	random_01	500x5000	network_simplex	3	209	206	211

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	random_01	500x5000	csa	3	1.4e+03	1.39e+03	1.41e+03
binary	random_01	500x5000	auction_scaled	3	2.48e+03	2.48e+03	2.5e+03
binary	random_05	500x500	auto	3	1.87	1.86	2
binary	random_05	500x500	hk01	3	1.88	1.66	2.34
binary	random_05	500x500	auction_scaled	3	2.7	2.66	3.02
binary	random_05	500x500	csa	3	3.32	3.14	3.86
binary	random_05	500x500	network_simplex	3	9.95	9.3	10.4
binary	random_05	500x500	csflow	3	25	24.7	25.2
binary	random_05	500x500	sap	3	25.2	25	25.2
binary	random_05	500x500	hungarian	3	57.4	57.3	57.6
binary	random_05	500x500	jv	3	57.9	57.7	58.3
binary	random_05	500x500	ramshaw_tarjan	3	58	57.9	58.6
binary	random_05	500x500	lapmod	3	62.4	62.3	62.5
binary	random_05	500x500	sap_dense	3	95	94.9	95.1
binary	random_05	500x1500	auto	3	5.24	4.61	5.26
binary	random_05	500x1500	hk01	3	5.32	5.29	5.73
binary	random_05	500x1500	network_simplex	3	34.2	33.4	34.4
binary	random_05	500x1500	csflow	3	48.4	48.4	49.3
binary	random_05	500x1500	sap	3	50.3	49.4	50.7
binary	random_05	500x1500	csa	3	86.2	85.8	87.1
binary	random_05	500x1500	auction_scaled	3	129	125	134
binary	random_05	500x1500	ramshaw_tarjan	3	178	178	180
binary	random_05	500x1500	hungarian	3	187	187	188
binary	random_05	500x1500	jv	3	187	186	187
binary	random_05	500x1500	lapmod	3	187	187	189
binary	random_05	500x1500	sap_dense	3	303	303	306
binary	random_05	500x5000	hk01	3	13.4	13.4	15.6
binary	random_05	500x5000	auto	3	17.8	15.7	17.8
binary	random_05	500x5000	sap	3	131	130	134
binary	random_05	500x5000	csflow	3	132	131	135
binary	random_05	500x5000	network_simplex	3	224	221	225
binary	random_05	500x5000	ramshaw_tarjan	3	577	572	584
binary	random_05	500x5000	lapmod	3	617	614	620
binary	random_05	500x5000	hungarian	3	634	632	636
binary	random_05	500x5000	jv	3	639	634	643
binary	random_05	500x5000	sap_dense	3	1.03e+03	1.03e+03	1.03e+03
binary	random_05	500x5000	csa	3	1.35e+03	1.29e+03	1.41e+03
binary	random_05	500x5000	auction_scaled	3	1.83e+03	1.82e+03	1.89e+03
binary	25%	500x500	hk01	3	3.14	2.93	3.14
binary	25%	500x500	auto	3	3.21	3.17	3.57
binary	25%	500x500	auction_scaled	3	6.11	6.1	6.33
binary	25%	500x500	csa	3	6.77	6.76	6.82
binary	25%	500x500	network_simplex	3	7.5	7.35	7.55
binary	25%	500x500	sap	3	45.3	45.3	45.3
binary	25%	500x500	csflow	3	45.7	45.4	45.8
binary	25%	500x500	jv	3	62.3	62.2	62.8
binary	25%	500x500	hungarian	3	62.7	62.5	62.8
binary	25%	500x500	lapmod	3	72	71.9	72.2
binary	25%	500x500	ramshaw_tarjan	3	101	101	101
binary	25%	500x500	sap_dense	3	126	126	126
binary	25%	500x1500	hk01	3	7.91	7.89	7.93

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	25%	500x1500	auto	3	8.15	8.05	8.64
binary	25%	500x1500	network_simplex	3	37.1	35.9	37.4
binary	25%	500x1500	csa	3	85.1	79.8	85.2
binary	25%	500x1500	sap	3	101	101	101
binary	25%	500x1500	csflow	3	102	101	102
binary	25%	500x1500	auction_scaled	3	109	106	115
binary	25%	500x1500	hungarian	3	205	205	205
binary	25%	500x1500	jv	3	206	205	206
binary	25%	500x1500	lapmod	3	222	222	222
binary	25%	500x1500	ramshaw_tarjan	3	389	389	390
binary	25%	500x1500	sap_dense	3	475	475	475
binary	25%	500x5000	hk01	3	26.7	26.6	28.5
binary	25%	500x5000	auto	3	30.4	30.4	30.4
binary	25%	500x5000	network_simplex	3	189	188	193
binary	25%	500x5000	csflow	3	298	298	302
binary	25%	500x5000	sap	3	300	299	300
binary	25%	500x5000	hungarian	3	700	700	702
binary	25%	500x5000	jv	3	704	704	704
binary	25%	500x5000	lapmod	3	740	740	742
binary	25%	500x5000	csa	3	1.11e+03	1.11e+03	1.23e+03
binary	25%	500x5000	ramshaw_tarjan	3	1.39e+03	1.39e+03	1.39e+03
binary	25%	500x5000	auction_scaled	3	1.67e+03	1.63e+03	1.69e+03
binary	25%	500x5000	sap_dense	3	1.7e+03	1.7e+03	1.7e+03
binary	60%	500x500	auto	3	4.45	4.36	4.45
binary	60%	500x500	hk01	3	4.46	4.45	4.47
binary	60%	500x500	network_simplex	3	7.19	7.02	7.26
binary	60%	500x500	auction_scaled	3	11.3	11.3	11.3
binary	60%	500x500	csa	3	11.8	11.7	11.8
binary	60%	500x500	jv	3	63.5	63.4	63.6
binary	60%	500x500	hungarian	3	63.6	63.3	64.2
binary	60%	500x500	sap	3	75.6	75.5	75.7
binary	60%	500x500	csflow	3	76.2	75.9	76.5
binary	60%	500x500	lapmod	3	84.4	84.1	84.9
binary	60%	500x500	sap_dense	3	131	130	131
binary	60%	500x500	ramshaw_tarjan	3	138	138	138
binary	60%	500x1500	auto	3	12.5	12.4	12.8
binary	60%	500x1500	hk01	3	12.9	12.7	13
binary	60%	500x1500	network_simplex	3	39.3	37.2	41.8
binary	60%	500x1500	csa	3	96.1	93.4	96.4
binary	60%	500x1500	auction_scaled	3	127	124	137
binary	60%	500x1500	sap	3	177	177	178
binary	60%	500x1500	csflow	3	179	179	179
binary	60%	500x1500	hungarian	3	206	205	207
binary	60%	500x1500	jv	3	207	207	208
binary	60%	500x1500	lapmod	3	260	259	260
binary	60%	500x1500	ramshaw_tarjan	3	535	534	536
binary	60%	500x1500	sap_dense	3	579	578	579
binary	60%	500x5000	auto	3	41.6	41.4	41.7
binary	60%	500x5000	hk01	3	43.7	42.6	43.7
binary	60%	500x5000	network_simplex	3	191	185	195
binary	60%	500x5000	csflow	3	569	566	582

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	60%	500x5000	sap	3	584	570	585
binary	60%	500x5000	jv	3	705	704	706
binary	60%	500x5000	hungarian	3	705	704	706
binary	60%	500x5000	lapmod	3	874	873	875
binary	60%	500x5000	csa	3	1.3e+03	1.22e+03	1.34e+03
binary	60%	500x5000	auction_scaled	3	1.65e+03	1.59e+03	1.66e+03
binary	60%	500x5000	ramshaw_tarjan	3	2.05e+03	1.99e+03	2.09e+03
binary	60%	500x5000	sap_dense	3	2.15e+03	2.15e+03	2.15e+03
constant	4 blocks	500x500	hk01	3	1.29	1.28	1.44
constant	4 blocks	500x500	auto	3	1.29	1.29	1.29
constant	4 blocks	500x500	network_simplex	3	3.43	3.41	3.56
constant	4 blocks	500x500	auction_scaled	3	3.44	3.43	3.46
constant	4 blocks	500x500	csa	3	3.67	3.6	4.11
constant	4 blocks	500x500	csflow	3	8.87	8.87	8.87
constant	4 blocks	500x500	sap	3	10.1	9.62	10.1
constant	4 blocks	500x500	ramshaw_tarjan	3	13.4	13.4	14.1
constant	4 blocks	500x500	hungarian	3	17.3	17.3	17.4
constant	4 blocks	500x500	jv	3	17.4	17.3	17.5
constant	4 blocks	500x500	lapmod	3	18.5	18.2	18.9
constant	4 blocks	500x500	sap_dense	3	25.6	25.3	25.8
constant	4 blocks	500x1500	hk01	3	3.59	3.58	4.07
constant	4 blocks	500x1500	auto	3	3.59	3.59	5.39
constant	4 blocks	500x1500	network_simplex	3	18.8	18.7	19.2
constant	4 blocks	500x1500	csflow	3	22.3	22.2	22.8
constant	4 blocks	500x1500	sap	3	22.5	22.3	22.8
constant	4 blocks	500x1500	ramshaw_tarjan	3	36.6	36.6	36.8
constant	4 blocks	500x1500	hungarian	3	52.7	52.7	52.7
constant	4 blocks	500x1500	jv	3	53.5	53	53.6
constant	4 blocks	500x1500	lapmod	3	56.8	56.8	56.8
constant	4 blocks	500x1500	sap_dense	3	75.8	75.7	75.9
constant	4 blocks	500x1500	csa	3	101	101	102
constant	4 blocks	500x1500	auction_scaled	3	150	150	150
constant	4 blocks	500x5000	auto	3	12.2	12.1	12.2
constant	4 blocks	500x5000	hk01	3	12.2	12.1	12.3
constant	4 blocks	500x5000	csflow	3	67.6	67.6	67.6
constant	4 blocks	500x5000	sap	3	73.2	70.1	73.8
constant	4 blocks	500x5000	network_simplex	3	103	103	103
constant	4 blocks	500x5000	ramshaw_tarjan	3	122	121	125
constant	4 blocks	500x5000	jv	3	178	177	179
constant	4 blocks	500x5000	hungarian	3	180	179	181
constant	4 blocks	500x5000	lapmod	3	190	188	191
constant	4 blocks	500x5000	sap_dense	3	255	253	255
constant	4 blocks	500x5000	csa	3	1.3e+03	1.3e+03	1.3e+03
constant	4 blocks	500x5000	auction_scaled	3	2.47e+03	2.46e+03	2.47e+03
constant	complete	500x500	hk01	3	1.76	1.76	1.77
constant	complete	500x500	auto	3	1.78	1.78	1.78
constant	complete	500x500	network_simplex	3	3.09	3.08	3.11
constant	complete	500x500	csa	3	8.94	8.93	8.94
constant	complete	500x500	auction_scaled	3	9.15	9.14	9.16
constant	complete	500x500	hungarian	3	57.7	57.7	59.3
constant	complete	500x500	jv	3	58.1	57.7	58.4

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
constant	complete	500x500	ramshaw_tarjan	3	58.9	58.8	58.9
constant	complete	500x500	sap_dense	3	88.2	88.1	88.5
constant	complete	500x500	lapmod	3	91.6	91.5	92
constant	complete	500x500	sap	3	98.6	98.6	98.9
constant	complete	500x500	csflow	3	99.8	99.8	99.9
constant	complete	500x1500	auto	3	4.97	4.97	5
constant	complete	500x1500	hk01	3	4.99	4.98	5
constant	complete	500x1500	network_simplex	3	9.09	9.08	9.12
constant	complete	500x1500	csa	3	77.7	77.3	77.8
constant	complete	500x1500	auction_scaled	3	185	185	185
constant	complete	500x1500	jv	3	195	195	196
constant	complete	500x1500	hungarian	3	196	196	197
constant	complete	500x1500	ramshaw_tarjan	3	200	200	201
constant	complete	500x1500	sap	3	239	239	240
constant	complete	500x1500	csflow	3	242	242	243
constant	complete	500x1500	sap_dense	3	278	278	280
constant	complete	500x1500	lapmod	3	287	287	287
constant	complete	500x5000	hk01	3	16.8	16.7	16.8
constant	complete	500x5000	auto	3	16.8	16.8	16.8
constant	complete	500x5000	network_simplex	3	30.5	30.5	30.5
constant	complete	500x5000	hungarian	3	669	669	671
constant	complete	500x5000	jv	3	673	671	673
constant	complete	500x5000	ramshaw_tarjan	3	755	754	802
constant	complete	500x5000	sap	3	760	760	760
constant	complete	500x5000	csflow	3	811	810	811
constant	complete	500x5000	csa	3	930	930	931
constant	complete	500x5000	sap_dense	3	952	951	952
constant	complete	500x5000	lapmod	3	981	981	981
constant	complete	500x5000	auction_scaled	3	2.31e+03	2.31e+03	2.31e+03
constant	random_01	500x500	hk01	3	1.15	1.15	1.17
constant	random_01	500x500	auto	3	1.6	1.38	1.6
constant	random_01	500x500	auction_scaled	3	2.02	1.97	2.39
constant	random_01	500x500	csa	3	2.45	2.4	3.03
constant	random_01	500x500	network_simplex	3	4.24	4.2	4.31
constant	random_01	500x500	csflow	3	10.4	10.3	10.5
constant	random_01	500x500	sap	3	10.8	10.5	10.9
constant	random_01	500x500	ramshaw_tarjan	3	35.1	35	35.7
constant	random_01	500x500	hungarian	3	43.2	43.1	44.2
constant	random_01	500x500	jv	3	43.4	43.3	44.4
constant	random_01	500x500	lapmod	3	47.5	47.4	48.3
constant	random_01	500x500	sap_dense	3	67.1	67	68.7
constant	random_01	500x1500	hk01	3	3.24	3.22	3.85
constant	random_01	500x1500	auto	3	4.47	3.84	4.49
constant	random_01	500x1500	csflow	3	17.5	17	17.8
constant	random_01	500x1500	sap	3	17.8	17.7	18.6
constant	random_01	500x1500	network_simplex	3	24.1	23.8	25.8
constant	random_01	500x1500	ramshaw_tarjan	3	86.2	83	87.9
constant	random_01	500x1500	csa	3	92	91.7	92.1
constant	random_01	500x1500	lapmod	3	123	118	125
constant	random_01	500x1500	hungarian	3	124	119	126
constant	random_01	500x1500	jv	3	124	118	126

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
constant	random_01	500x1500	auction_scaled	3	156	152	157
constant	random_01	500x1500	sap_dense	3	183	176	186
constant	random_01	500x5000	hk01	3	12	11.8	14.1
constant	random_01	500x5000	auto	3	15.9	15.9	15.9
constant	random_01	500x5000	sap	3	40.6	39.9	41.1
constant	random_01	500x5000	csflow	3	44.8	42.9	44.8
constant	random_01	500x5000	network_simplex	3	149	149	149
constant	random_01	500x5000	ramshaw_tarjan	3	242	237	247
constant	random_01	500x5000	lapmod	3	354	346	360
constant	random_01	500x5000	hungarian	3	371	364	375
constant	random_01	500x5000	jv	3	372	364	377
constant	random_01	500x5000	sap_dense	3	544	535	559
constant	random_01	500x5000	csa	3	1.52e+03	1.51e+03	1.52e+03
constant	random_01	500x5000	auction_scaled	3	2.5e+03	2.35e+03	2.52e+03
constant	random_05	500x500	hk01	3	1.35	1.35	1.35
constant	random_05	500x500	auto	3	1.36	1.36	1.37
constant	random_05	500x500	csa	3	2.74	2.73	2.95
constant	random_05	500x500	auction_scaled	3	3.19	3.11	3.54
constant	random_05	500x500	network_simplex	3	3.85	3.8	4.51
constant	random_05	500x500	csflow	3	18.6	18.5	18.7
constant	random_05	500x500	sap	3	19	18.8	19.6
constant	random_05	500x500	ramshaw_tarjan	3	53.5	53.4	53.6
constant	random_05	500x500	hungarian	3	59.9	59.6	59.9
constant	random_05	500x500	jv	3	60	59.8	60
constant	random_05	500x500	lapmod	3	63.1	63.1	63.3
constant	random_05	500x500	sap_dense	3	97.1	96.8	97.3
constant	random_05	500x1500	hk01	3	3.72	3.71	3.73
constant	random_05	500x1500	auto	3	5.01	4.37	5.02
constant	random_05	500x1500	network_simplex	3	24.2	23.5	26.5
constant	random_05	500x1500	csflow	3	36.8	36.7	37.3
constant	random_05	500x1500	sap	3	38	37.8	38.1
constant	random_05	500x1500	csa	3	80.3	76.4	80.9
constant	random_05	500x1500	auction_scaled	3	142	140	142
constant	random_05	500x1500	ramshaw_tarjan	3	171	171	172
constant	random_05	500x1500	lapmod	3	195	194	195
constant	random_05	500x1500	jv	3	196	196	197
constant	random_05	500x1500	hungarian	3	198	196	198
constant	random_05	500x1500	sap_dense	3	315	314	315
constant	random_05	500x5000	hk01	3	12.5	12.5	14.7
constant	random_05	500x5000	auto	3	16.9	14.8	16.9
constant	random_05	500x5000	csflow	3	101	98.6	101
constant	random_05	500x5000	sap	3	101	99.3	101
constant	random_05	500x5000	network_simplex	3	141	139	145
constant	random_05	500x5000	ramshaw_tarjan	3	577	573	577
constant	random_05	500x5000	lapmod	3	646	645	649
constant	random_05	500x5000	hungarian	3	670	669	671
constant	random_05	500x5000	jv	3	673	673	674
constant	random_05	500x5000	sap_dense	3	1.08e+03	1.08e+03	1.08e+03
constant	random_05	500x5000	csa	3	1.27e+03	1.25e+03	1.33e+03
constant	random_05	500x5000	auction_scaled	3	2.07e+03	2.02e+03	2.12e+03
constant	25%	500x500	hk01	3	2.3	2.3	2.49

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
constant	25%	500x500	auto	3	2.71	2.7	3.11
constant	25%	500x500	network_simplex	3	4.79	4.63	5.22
constant	25%	500x500	auction_scaled	3	5.54	5.32	5.71
constant	25%	500x500	csa	3	5.74	5.73	5.88
constant	25%	500x500	csflow	3	37.8	37.6	37.9
constant	25%	500x500	sap	3	38.7	38.2	38.8
constant	25%	500x500	hungarian	3	62.1	62.1	62.8
constant	25%	500x500	jv	3	62.9	62.7	63.2
constant	25%	500x500	lapmod	3	70.2	70	70.2
constant	25%	500x500	ramshaw_tarjan	3	93.8	93.6	94.5
constant	25%	500x500	sap_dense	3	124	124	124
constant	25%	500x1500	hk01	3	6.52	6.51	7.08
constant	25%	500x1500	auto	3	7.63	7.62	7.65
constant	25%	500x1500	network_simplex	3	25.8	25.7	27.4
constant	25%	500x1500	csa	3	65.5	64.4	65.6
constant	25%	500x1500	sap	3	83.6	83.3	83.8
constant	25%	500x1500	csflow	3	84.6	84	84.7
constant	25%	500x1500	auction_scaled	3	130	127	135
constant	25%	500x1500	jv	3	204	204	204
constant	25%	500x1500	hungarian	3	204	203	205
constant	25%	500x1500	lapmod	3	219	219	220
constant	25%	500x1500	ramshaw_tarjan	3	377	376	377
constant	25%	500x1500	sap_dense	3	470	470	471
constant	25%	500x5000	hk01	3	25.5	25.5	25.5
constant	25%	500x5000	auto	3	25.5	25.5	25.6
constant	25%	500x5000	network_simplex	3	134	133	134
constant	25%	500x5000	sap	3	251	251	254
constant	25%	500x5000	csflow	3	254	253	255
constant	25%	500x5000	hungarian	3	694	693	695
constant	25%	500x5000	jv	3	695	694	695
constant	25%	500x5000	lapmod	3	733	733	734
constant	25%	500x5000	csa	3	973	886	1e+03
constant	25%	500x5000	ramshaw_tarjan	3	1.35e+03	1.35e+03	1.35e+03
constant	25%	500x5000	sap_dense	3	1.69e+03	1.69e+03	1.69e+03
constant	25%	500x5000	auction_scaled	3	1.88e+03	1.88e+03	2.03e+03
constant	60%	500x500	hk01	3	3.31	3.26	3.38
constant	60%	500x500	auto	3	3.42	3.37	3.43
constant	60%	500x500	network_simplex	3	5.19	5.09	5.21
constant	60%	500x500	auction_scaled	3	9.21	8.77	9.65
constant	60%	500x500	csa	3	9.38	8.94	9.85
constant	60%	500x500	hungarian	3	61.8	61.7	62.1
constant	60%	500x500	jv	3	63.1	62.6	63.3
constant	60%	500x500	sap	3	68.2	68.2	68.2
constant	60%	500x500	csflow	3	69.6	69.3	69.9
constant	60%	500x500	lapmod	3	82.6	82	82.8
constant	60%	500x500	sap_dense	3	130	129	130
constant	60%	500x500	ramshaw_tarjan	3	131	131	131
constant	60%	500x1500	hk01	3	8.97	8.94	9.29
constant	60%	500x1500	auto	3	9.61	9.3	9.61
constant	60%	500x1500	network_simplex	3	29.5	27.4	29.7
constant	60%	500x1500	csa	3	84.1	79.4	84.3

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
constant	60%	500x1500	sap	3	161	161	161
constant	60%	500x1500	csflow	3	163	163	163
constant	60%	500x1500	auction_scaled	3	164	138	171
constant	60%	500x1500	hungarian	3	203	203	204
constant	60%	500x1500	jv	3	204	204	205
constant	60%	500x1500	lapmod	3	256	256	257
constant	60%	500x1500	ramshaw_tarjan	3	524	522	525
constant	60%	500x1500	sap_dense	3	574	574	575
constant	60%	500x5000	hk01	3	32.2	31.2	32.2
constant	60%	500x5000	auto	3	32.5	31.3	36
constant	60%	500x5000	network_simplex	3	110	110	121
constant	60%	500x5000	sap	3	507	507	524
constant	60%	500x5000	csflow	3	544	537	545
constant	60%	500x5000	jv	3	693	693	694
constant	60%	500x5000	hungarian	3	694	693	695
constant	60%	500x5000	lapmod	3	864	863	864
constant	60%	500x5000	csa	3	1.05e+03	1.01e+03	1.09e+03
constant	60%	500x5000	ramshaw_tarjan	3	2.01e+03	1.94e+03	2.02e+03
constant	60%	500x5000	sap_dense	3	2.13e+03	2.13e+03	2.14e+03
constant	60%	500x5000	auction_scaled	3	2.15e+03	2.04e+03	2.25e+03
dbl_uniform	4 blocks	500x500	auction_scaled	3	2.93	2.93	2.99
dbl_uniform	4 blocks	500x500	csa	3	3.74	3.61	3.76
dbl_uniform	4 blocks	500x500	hungarian	3	3.92	3.8	4.01
dbl_uniform	4 blocks	500x500	jv	3	4.01	3.78	4.03
dbl_uniform	4 blocks	500x500	auto	3	4.08	3.82	4.49
dbl_uniform	4 blocks	500x500	lapmod	3	4.57	4.34	4.61
dbl_uniform	4 blocks	500x500	csflow	3	5.13	5.09	5.15
dbl_uniform	4 blocks	500x500	ramshaw_tarjan	3	5.23	5.15	5.29
dbl_uniform	4 blocks	500x500	sap	3	5.42	5.28	5.48
dbl_uniform	4 blocks	500x500	sap_dense	3	6.76	6.68	6.83
dbl_uniform	4 blocks	500x500	network_simplex	3	30	29.2	30
dbl_uniform	4 blocks	500x1500	hungarian	3	3.75	3.72	3.79
dbl_uniform	4 blocks	500x1500	jv	3	3.76	3.73	3.79
dbl_uniform	4 blocks	500x1500	lapmod	3	4.33	4.3	4.38
dbl_uniform	4 blocks	500x1500	auto	3	4.71	4.23	5.31
dbl_uniform	4 blocks	500x1500	ramshaw_tarjan	3	8.46	7.96	9.78
dbl_uniform	4 blocks	500x1500	csflow	3	9.67	9.63	9.78
dbl_uniform	4 blocks	500x1500	sap_dense	3	9.84	9.83	9.96
dbl_uniform	4 blocks	500x1500	sap	3	10.9	10.3	12.2
dbl_uniform	4 blocks	500x1500	network_simplex	3	99.7	98.4	102
dbl_uniform	4 blocks	500x1500	auction_scaled	3	279	267	286
dbl_uniform	4 blocks	500x1500	csa	3	508	499	565
dbl_uniform	4 blocks	500x5000	auto	3	12.1	12.1	13.8
dbl_uniform	4 blocks	500x5000	jv	3	12.2	12.1	13.8
dbl_uniform	4 blocks	500x5000	hungarian	3	15.1	13.6	15.1
dbl_uniform	4 blocks	500x5000	lapmod	3	16.9	15.2	20.9
dbl_uniform	4 blocks	500x5000	ramshaw_tarjan	3	27.1	25.5	27.2
dbl_uniform	4 blocks	500x5000	sap_dense	3	28.9	28.9	31.3
dbl_uniform	4 blocks	500x5000	csflow	3	30.9	30.8	31
dbl_uniform	4 blocks	500x5000	sap	3	35	33.4	35.2
dbl_uniform	4 blocks	500x5000	network_simplex	3	561	554	573

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	4 blocks	500x5000	auction_scaled	3	3.74e+03	3.69e+03	3.77e+03
dbl_uniform	4 blocks	500x5000	csa	3	1.08e+04	1.08e+04	1.11e+04
dbl_uniform	complete	500x500	jv	3	3.19	3.16	3.74
dbl_uniform	complete	500x500	auto	3	3.2	3.17	3.83
dbl_uniform	complete	500x500	hungarian	3	7.7	7.68	8.01
dbl_uniform	complete	500x500	auction_scaled	3	8.15	8.08	8.19
dbl_uniform	complete	500x500	csa	3	10.4	10.3	10.5
dbl_uniform	complete	500x500	lapmod	3	10.8	10.8	11.3
dbl_uniform	complete	500x500	sap_dense	3	11.4	11.3	11.8
dbl_uniform	complete	500x500	ramshaw_tarjan	3	13.3	13.3	13.7
dbl_uniform	complete	500x500	sap	3	19.7	19.7	20.4
dbl_uniform	complete	500x500	csflow	3	19.7	19.7	20.1
dbl_uniform	complete	500x500	network_simplex	3	96.9	89.2	103
dbl_uniform	complete	500x1500	hungarian	3	3.96	3.87	4.03
dbl_uniform	complete	500x1500	auto	3	4.11	3.97	4.12
dbl_uniform	complete	500x1500	jv	3	4.11	4.04	5.06
dbl_uniform	complete	500x1500	lapmod	3	5.39	5.36	5.46
dbl_uniform	complete	500x1500	sap_dense	3	9.47	9.27	9.57
dbl_uniform	complete	500x1500	ramshaw_tarjan	3	14.3	14.2	14.4
dbl_uniform	complete	500x1500	csflow	3	27.3	27.1	27.6
dbl_uniform	complete	500x1500	sap	3	27.3	27.1	27.6
dbl_uniform	complete	500x1500	network_simplex	3	101	100	126
dbl_uniform	complete	500x1500	auction_scaled	3	266	251	273
dbl_uniform	complete	500x1500	csa	3	507	494	544
dbl_uniform	complete	500x5000	hungarian	3	11.5	11.4	11.7
dbl_uniform	complete	500x5000	jv	3	11.5	11.5	11.7
dbl_uniform	complete	500x5000	auto	3	11.5	11.5	11.7
dbl_uniform	complete	500x5000	lapmod	3	22.1	19.5	22.4
dbl_uniform	complete	500x5000	sap_dense	3	25.5	25.3	26
dbl_uniform	complete	500x5000	ramshaw_tarjan	3	49.2	47.9	52.7
dbl_uniform	complete	500x5000	csflow	3	102	101	102
dbl_uniform	complete	500x5000	sap	3	103	102	103
dbl_uniform	complete	500x5000	network_simplex	3	569	569	594
dbl_uniform	complete	500x5000	auction_scaled	3	3.22e+03	3.21e+03	3.26e+03
dbl_uniform	complete	500x5000	csa	3	1.01e+04	9.9e+03	1.06e+04
dbl_uniform	random_01	500x500	auction_scaled	3	1.73	1.55	1.75
dbl_uniform	random_01	500x500	csa	3	2.13	2.07	2.6
dbl_uniform	random_01	500x500	csflow	3	3.07	2.95	3.16
dbl_uniform	random_01	500x500	sap	3	3.32	3.29	3.79
dbl_uniform	random_01	500x500	auto	3	6.36	6.1	6.89
dbl_uniform	random_01	500x500	lapmod	3	6.39	6.08	6.42
dbl_uniform	random_01	500x500	jv	3	6.42	6.42	6.57
dbl_uniform	random_01	500x500	ramshaw_tarjan	3	6.43	6.36	7.07
dbl_uniform	random_01	500x500	hungarian	3	6.47	6.44	6.59
dbl_uniform	random_01	500x500	sap_dense	3	10.2	9.6	10.2
dbl_uniform	random_01	500x500	network_simplex	3	45.3	45.1	48
dbl_uniform	random_01	500x1500	hungarian	3	3.73	3.72	3.75
dbl_uniform	random_01	500x1500	csflow	3	3.82	3.81	4.46
dbl_uniform	random_01	500x1500	lapmod	3	4.1	4.09	4.73
dbl_uniform	random_01	500x1500	auto	3	5.01	5.01	5.41
dbl_uniform	random_01	500x1500	jv	3	5.01	4.41	5.02

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	random_01	500x1500	sap	3	5.23	4.58	5.25
dbl_uniform	random_01	500x1500	ramshaw_tarjan	3	6.95	6.31	6.96
dbl_uniform	random_01	500x1500	sap_dense	3	10.1	9.84	10.6
dbl_uniform	random_01	500x1500	network_simplex	3	73.2	71.1	76.3
dbl_uniform	random_01	500x1500	auction_scaled	3	316	311	321
dbl_uniform	random_01	500x1500	csa	3	575	543	617
dbl_uniform	random_01	500x5000	auto	3	12.6	12.5	14.7
dbl_uniform	random_01	500x5000	csflow	3	13.5	13.5	14.6
dbl_uniform	random_01	500x5000	hungarian	3	16.5	14.5	19.9
dbl_uniform	random_01	500x5000	jv	3	16.6	14.4	16.7
dbl_uniform	random_01	500x5000	lapmod	3	17.9	15.8	17.9
dbl_uniform	random_01	500x5000	sap	3	18.2	16.3	18.4
dbl_uniform	random_01	500x5000	ramshaw_tarjan	3	23.8	23.8	23.9
dbl_uniform	random_01	500x5000	sap_dense	3	33.8	32.4	34.2
dbl_uniform	random_01	500x5000	network_simplex	3	320	316	323
dbl_uniform	random_01	500x5000	auction_scaled	3	4.46e+03	4.16e+03	4.71e+03
dbl_uniform	random_01	500x5000	csa	3	1.21e+04	1.14e+04	1.22e+04
dbl_uniform	random_05	500x500	auction_scaled	3	2.32	2.3	2.33
dbl_uniform	random_05	500x500	csa	3	2.71	2.7	2.76
dbl_uniform	random_05	500x500	sap	3	5.43	5.2	5.72
dbl_uniform	random_05	500x500	csflow	3	5.85	5.41	6.15
dbl_uniform	random_05	500x500	jv	3	6.96	6.69	7.42
dbl_uniform	random_05	500x500	auto	3	7.36	6.86	8.05
dbl_uniform	random_05	500x500	hungarian	3	7.38	7.11	7.64
dbl_uniform	random_05	500x500	lapmod	3	7.46	7.13	7.68
dbl_uniform	random_05	500x500	ramshaw_tarjan	3	10	9.62	10.4
dbl_uniform	random_05	500x500	sap_dense	3	13.5	12.5	13.5
dbl_uniform	random_05	500x500	network_simplex	3	90.3	89.8	93.6
dbl_uniform	random_05	500x1500	lapmod	3	4.6	4.59	4.63
dbl_uniform	random_05	500x1500	csflow	3	5.29	5.27	5.9
dbl_uniform	random_05	500x1500	auto	3	5.4	4.8	5.42
dbl_uniform	random_05	500x1500	hungarian	3	5.4	5.39	5.42
dbl_uniform	random_05	500x1500	jv	3	5.44	5.41	6.81
dbl_uniform	random_05	500x1500	ramshaw_tarjan	3	6.67	6.63	7.31
dbl_uniform	random_05	500x1500	sap	3	9.35	7.43	9.5
dbl_uniform	random_05	500x1500	sap_dense	3	14	12.4	14.6
dbl_uniform	random_05	500x1500	network_simplex	3	105	102	114
dbl_uniform	random_05	500x1500	auction_scaled	3	270	257	275
dbl_uniform	random_05	500x1500	csa	3	491	442	542
dbl_uniform	random_05	500x5000	csflow	3	17.5	17.5	17.6
dbl_uniform	random_05	500x5000	hungarian	3	17.7	17.7	17.8
dbl_uniform	random_05	500x5000	jv	3	17.8	15.7	17.8
dbl_uniform	random_05	500x5000	auto	3	17.9	17.8	17.9
dbl_uniform	random_05	500x5000	lapmod	3	19.3	17.2	19.4
dbl_uniform	random_05	500x5000	sap	3	22.3	20.1	22.3
dbl_uniform	random_05	500x5000	ramshaw_tarjan	3	25.9	23.9	26
dbl_uniform	random_05	500x5000	sap_dense	3	32.2	32.2	32.5
dbl_uniform	random_05	500x5000	network_simplex	3	476	476	562
dbl_uniform	random_05	500x5000	auction_scaled	3	4.11e+03	3.8e+03	4.13e+03
dbl_uniform	random_05	500x5000	csa	3	1.04e+04	1.02e+04	1.13e+04
dbl_uniform	25%	500x500	auction_scaled	3	4.59	4.57	5.06

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	25%	500x500	csa	3	5.75	5.74	6.23
dbl_uniform	25%	500x500	hungarian	3	8.98	8.66	9.56
dbl_uniform	25%	500x500	jv	3	8.99	8.65	9.48
dbl_uniform	25%	500x500	auto	3	9	8.63	9.5
dbl_uniform	25%	500x500	sap	3	10.4	10.2	10.6
dbl_uniform	25%	500x500	csflow	3	10.7	10.3	10.9
dbl_uniform	25%	500x500	lapmod	3	10.9	10.4	11.9
dbl_uniform	25%	500x500	ramshaw_tarjan	3	18.1	17.2	19.3
dbl_uniform	25%	500x500	sap_dense	3	18.9	18	20
dbl_uniform	25%	500x500	network_simplex	3	91	86.9	97
dbl_uniform	25%	500x1500	auto	3	7.36	6.87	7.39
dbl_uniform	25%	500x1500	jv	3	7.4	7.36	8.63
dbl_uniform	25%	500x1500	hungarian	3	7.42	6.82	7.88
dbl_uniform	25%	500x1500	lapmod	3	8.3	8.26	9.56
dbl_uniform	25%	500x1500	ramshaw_tarjan	3	11.5	11.4	11.5
dbl_uniform	25%	500x1500	csflow	3	12.7	12.7	13.2
dbl_uniform	25%	500x1500	sap	3	13.9	13.9	14.3
dbl_uniform	25%	500x1500	sap_dense	3	13.9	13.6	14.5
dbl_uniform	25%	500x1500	network_simplex	3	117	113	162
dbl_uniform	25%	500x1500	auction_scaled	3	238	237	242
dbl_uniform	25%	500x1500	csa	3	577	530	607
dbl_uniform	25%	500x5000	jv	3	19.8	19.8	19.9
dbl_uniform	25%	500x5000	hungarian	3	23.6	23.5	23.6
dbl_uniform	25%	500x5000	auto	3	23.6	21.7	23.6
dbl_uniform	25%	500x5000	lapmod	3	27.1	25.4	27.2
dbl_uniform	25%	500x5000	ramshaw_tarjan	3	36.9	36.8	38.5
dbl_uniform	25%	500x5000	sap	3	41.7	41.6	43.8
dbl_uniform	25%	500x5000	csflow	3	42.3	42.1	42.4
dbl_uniform	25%	500x5000	sap_dense	3	46.9	44.5	50.3
dbl_uniform	25%	500x5000	network_simplex	3	615	612	631
dbl_uniform	25%	500x5000	auction_scaled	3	3.4e+03	3.39e+03	3.43e+03
dbl_uniform	25%	500x5000	csa	3	1.03e+04	1e+04	1.09e+04
dbl_uniform	60%	500x500	auction_scaled	3	7.69	7.47	7.84
dbl_uniform	60%	500x500	jv	3	9.16	8.81	9.86
dbl_uniform	60%	500x500	auto	3	9.19	8.71	9.86
dbl_uniform	60%	500x500	hungarian	3	9.22	8.84	9.88
dbl_uniform	60%	500x500	csa	3	10.1	9.89	10.6
dbl_uniform	60%	500x500	lapmod	3	12.8	12.1	14.2
dbl_uniform	60%	500x500	csflow	3	16.5	16.5	17.2
dbl_uniform	60%	500x500	sap	3	16.6	16.3	17.1
dbl_uniform	60%	500x500	sap_dense	3	20.6	19.5	22.4
dbl_uniform	60%	500x500	ramshaw_tarjan	3	22.3	21.3	24.1
dbl_uniform	60%	500x500	network_simplex	3	104	91.5	105
dbl_uniform	60%	500x1500	jv	3	8.46	8.28	8.48
dbl_uniform	60%	500x1500	hungarian	3	8.51	8.51	8.69
dbl_uniform	60%	500x1500	auto	3	8.54	8.53	8.61
dbl_uniform	60%	500x1500	lapmod	3	10.2	10.1	10.3
dbl_uniform	60%	500x1500	sap_dense	3	17.4	16.9	17.5
dbl_uniform	60%	500x1500	ramshaw_tarjan	3	18	17.9	18
dbl_uniform	60%	500x1500	csflow	3	22.9	22.9	23.2
dbl_uniform	60%	500x1500	sap	3	23.9	23.7	24.9

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	60%	500x1500	network_simplex	3	124	123	137
dbl_uniform	60%	500x1500	auction_scaled	3	271	267	285
dbl_uniform	60%	500x1500	csa	3	580	550	781
dbl_uniform	60%	500x5000	auto	3	24.9	24.9	25
dbl_uniform	60%	500x5000	hungarian	3	26.9	25.8	27
dbl_uniform	60%	500x5000	jv	3	27	26	27.1
dbl_uniform	60%	500x5000	lapmod	3	33.4	33.3	38.1
dbl_uniform	60%	500x5000	sap_dense	3	52.4	51.4	52.7
dbl_uniform	60%	500x5000	ramshaw_tarjan	3	57.8	57.4	57.9
dbl_uniform	60%	500x5000	csflow	3	74.5	74.4	75.3
dbl_uniform	60%	500x5000	sap	3	76	75.4	76.6
dbl_uniform	60%	500x5000	network_simplex	3	715	677	731
dbl_uniform	60%	500x5000	auction_scaled	3	3.19e+03	2.99e+03	3.33e+03
dbl_uniform	60%	500x5000	csa	3	1.01e+04	1e+04	1.02e+04
heavy_tailed	4 blocks	500x500	hungarian	3	4.25	4.15	4.53
heavy_tailed	4 blocks	500x500	auto	3	4.47	4.24	4.53
heavy_tailed	4 blocks	500x500	jv	3	4.48	4.24	4.53
heavy_tailed	4 blocks	500x500	lapmod	3	4.98	4.81	5.08
heavy_tailed	4 blocks	500x500	csflow	3	5.5	5.46	5.71
heavy_tailed	4 blocks	500x500	sap	3	5.56	5.5	5.72
heavy_tailed	4 blocks	500x500	ramshaw_tarjan	3	5.84	5.6	5.92
heavy_tailed	4 blocks	500x500	auction_scaled	3	6.17	6.11	6.23
heavy_tailed	4 blocks	500x500	csa	3	7.17	7.12	7.3
heavy_tailed	4 blocks	500x500	sap_dense	3	7.19	6.99	7.34
heavy_tailed	4 blocks	500x500	network_simplex	3	30.6	29	31.6
heavy_tailed	4 blocks	500x1500	jv	3	3.69	3.69	3.74
heavy_tailed	4 blocks	500x1500	hungarian	3	3.71	3.69	3.74
heavy_tailed	4 blocks	500x1500	auto	3	3.71	3.7	3.73
heavy_tailed	4 blocks	500x1500	lapmod	3	4.29	4.28	4.75
heavy_tailed	4 blocks	500x1500	ramshaw_tarjan	3	7.49	7.47	7.55
heavy_tailed	4 blocks	500x1500	sap_dense	3	8.18	8.16	8.24
heavy_tailed	4 blocks	500x1500	csflow	3	9.65	9.62	10.2
heavy_tailed	4 blocks	500x1500	sap	3	9.93	9.93	10.3
heavy_tailed	4 blocks	500x1500	network_simplex	3	114	107	114
heavy_tailed	4 blocks	500x1500	csa	3	185	183	204
heavy_tailed	4 blocks	500x1500	auction_scaled	1	264	264	264
heavy_tailed	4 blocks	500x5000	hungarian	3	12.2	12.1	13.7
heavy_tailed	4 blocks	500x5000	jv	3	12.3	12.2	13.8
heavy_tailed	4 blocks	500x5000	auto	3	12.4	12.2	13.9
heavy_tailed	4 blocks	500x5000	lapmod	3	16.9	15.3	18.1
heavy_tailed	4 blocks	500x5000	ramshaw_tarjan	3	27.2	26	27.4
heavy_tailed	4 blocks	500x5000	sap_dense	3	29.7	29.6	30.8
heavy_tailed	4 blocks	500x5000	csflow	3	31.5	31.1	32.6
heavy_tailed	4 blocks	500x5000	sap	3	32.1	31.8	33.3
heavy_tailed	4 blocks	500x5000	network_simplex	3	679	663	692
heavy_tailed	4 blocks	500x5000	csa	3	7.43e+03	7.38e+03	7.44e+03
heavy_tailed	complete	500x500	jv	3	3.62	3.62	3.83
heavy_tailed	complete	500x500	auto	3	3.63	3.61	3.82
heavy_tailed	complete	500x500	hungarian	3	8.86	8.02	9.07
heavy_tailed	complete	500x500	lapmod	3	11.9	11.1	13
heavy_tailed	complete	500x500	sap_dense	3	12	11.3	12.8

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
heavy_tailed	complete	500x500	ramshaw_tarjan	3	13.7	13.1	14.4
heavy_tailed	complete	500x500	csa	3	19.1	18.2	19.2
heavy_tailed	complete	500x500	csflow	3	20.1	19.5	21
heavy_tailed	complete	500x500	sap	3	20.1	19.5	20.9
heavy_tailed	complete	500x500	auction_scaled	3	20.5	20.1	20.9
heavy_tailed	complete	500x500	network_simplex	3	92.2	90.7	97.1
heavy_tailed	complete	500x1500	jv	3	3.77	3.76	3.82
heavy_tailed	complete	500x1500	auto	3	3.78	3.75	3.82
heavy_tailed	complete	500x1500	hungarian	3	3.8	3.79	3.83
heavy_tailed	complete	500x1500	lapmod	3	5.26	5.23	5.33
heavy_tailed	complete	500x1500	sap_dense	3	7.26	7.23	7.3
heavy_tailed	complete	500x1500	ramshaw_tarjan	3	14	14	14.1
heavy_tailed	complete	500x1500	csflow	3	27.2	27.1	27.2
heavy_tailed	complete	500x1500	sap	3	27.2	27.2	27.3
heavy_tailed	complete	500x1500	network_simplex	3	119	112	136
heavy_tailed	complete	500x1500	csa	3	149	147	150
heavy_tailed	complete	500x1500	auction_scaled	1	258	258	258
heavy_tailed	complete	500x5000	hungarian	3	11.3	11.2	11.3
heavy_tailed	complete	500x5000	auto	3	11.4	11.4	11.5
heavy_tailed	complete	500x5000	jv	3	11.5	11.5	11.7
heavy_tailed	complete	500x5000	lapmod	3	22.5	22.3	22.6
heavy_tailed	complete	500x5000	sap_dense	3	27.2	26.8	27.5
heavy_tailed	complete	500x5000	ramshaw_tarjan	3	48.5	48.3	49.5
heavy_tailed	complete	500x5000	sap	3	102	102	103
heavy_tailed	complete	500x5000	csflow	3	104	102	105
heavy_tailed	complete	500x5000	network_simplex	3	793	748	815
heavy_tailed	complete	500x5000	csa	3	7.08e+03	6.96e+03	7.09e+03
heavy_tailed	random_01	500x500	auction_scaled	3	2.35	2.29	2.5
heavy_tailed	random_01	500x500	csa	3	2.67	2.62	2.71
heavy_tailed	random_01	500x500	sap	3	3.68	3.46	3.82
heavy_tailed	random_01	500x500	csflow	3	3.69	3.65	3.81
heavy_tailed	random_01	500x500	auto	3	7.58	7.45	8.13
heavy_tailed	random_01	500x500	lapmod	3	7.66	7.53	7.95
heavy_tailed	random_01	500x500	hungarian	3	7.72	7.63	8.21
heavy_tailed	random_01	500x500	jv	3	7.77	7.66	8.24
heavy_tailed	random_01	500x500	ramshaw_tarjan	3	8.15	8.09	8.72
heavy_tailed	random_01	500x500	sap_dense	3	12.2	12.1	12.9
heavy_tailed	random_01	500x500	network_simplex	3	39.7	39.6	40.9
heavy_tailed	random_01	500x1500	hungarian	3	3.73	3.73	4.39
heavy_tailed	random_01	500x1500	auto	3	3.8	3.78	4.4
heavy_tailed	random_01	500x1500	csflow	3	3.83	3.81	3.9
heavy_tailed	random_01	500x1500	jv	3	5.01	4.38	5.01
heavy_tailed	random_01	500x1500	lapmod	3	5.37	4.99	5.39
heavy_tailed	random_01	500x1500	sap	3	6.47	5.23	7.25
heavy_tailed	random_01	500x1500	ramshaw_tarjan	3	6.92	6.32	6.92
heavy_tailed	random_01	500x1500	sap_dense	3	8.24	8.21	8.84
heavy_tailed	random_01	500x1500	network_simplex	3	81.7	74	89.2
heavy_tailed	random_01	500x1500	auction_scaled	3	368	360	389
heavy_tailed	random_01	500x1500	csa	3	372	366	405
heavy_tailed	random_01	500x5000	hungarian	3	12.3	12.2	14.4
heavy_tailed	random_01	500x5000	sap	3	14.2	14.2	16.4

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
heavy_tailed	random_01	500x5000	auto	3	16.5	14.4	16.5
heavy_tailed	random_01	500x5000	jv	3	16.5	14.5	16.6
heavy_tailed	random_01	500x5000	lapmod	3	17.7	15.7	17.7
heavy_tailed	random_01	500x5000	csflow	3	17.9	16.7	18
heavy_tailed	random_01	500x5000	ramshaw_tarjan	3	19.5	19.4	19.5
heavy_tailed	random_01	500x5000	sap_dense	3	29	26.9	30.4
heavy_tailed	random_01	500x5000	network_simplex	3	321	319	322
heavy_tailed	random_01	500x5000	csa	3	7.93e+03	7.74e+03	7.94e+03
heavy_tailed	random_05	500x500	auction_scaled	3	3.43	3.39	3.58
heavy_tailed	random_05	500x500	csa	3	5.15	4.36	5.38
heavy_tailed	random_05	500x500	csflow	3	5.29	5.25	5.48
heavy_tailed	random_05	500x500	sap	3	5.43	5.15	5.77
heavy_tailed	random_05	500x500	auto	3	7.94	7.78	9.03
heavy_tailed	random_05	500x500	jv	3	7.95	7.8	9.01
heavy_tailed	random_05	500x500	hungarian	3	7.97	7.59	9.06
heavy_tailed	random_05	500x500	lapmod	3	8.14	7.96	9.79
heavy_tailed	random_05	500x500	ramshaw_tarjan	3	10.9	10.6	12.3
heavy_tailed	random_05	500x500	sap_dense	3	13.7	13.3	15.4
heavy_tailed	random_05	500x500	network_simplex	3	74.4	74.1	75.5
heavy_tailed	random_05	500x1500	lapmod	3	4.58	4.57	5.22
heavy_tailed	random_05	500x1500	jv	3	5.37	4.74	5.39
heavy_tailed	random_05	500x1500	auto	3	5.38	4.76	5.4
heavy_tailed	random_05	500x1500	hungarian	3	5.42	4.76	6.76
heavy_tailed	random_05	500x1500	csflow	3	6.52	5.89	6.54
heavy_tailed	random_05	500x1500	ramshaw_tarjan	3	6.66	6.62	7.31
heavy_tailed	random_05	500x1500	sap	3	6.68	6.65	6.69
heavy_tailed	random_05	500x1500	sap_dense	3	10	10	10
heavy_tailed	random_05	500x1500	network_simplex	3	106	97.3	110
heavy_tailed	random_05	500x1500	csa	3	171	155	172
heavy_tailed	random_05	500x1500	auction_scaled	2	287	282	291
heavy_tailed	random_05	500x5000	hungarian	3	14	13.8	15.8
heavy_tailed	random_05	500x5000	auto	3	17.7	15.6	17.7
heavy_tailed	random_05	500x5000	jv	3	17.7	15.7	17.8
heavy_tailed	random_05	500x5000	sap	3	18.1	18	19.4
heavy_tailed	random_05	500x5000	lapmod	3	19.3	17.9	19.3
heavy_tailed	random_05	500x5000	ramshaw_tarjan	3	21.8	21.8	23.9
heavy_tailed	random_05	500x5000	csflow	3	21.8	19.7	21.8
heavy_tailed	random_05	500x5000	sap_dense	3	32.8	31.9	33.3
heavy_tailed	random_05	500x5000	network_simplex	3	536	530	570
heavy_tailed	random_05	500x5000	csa	3	7e+03	6.94e+03	7.1e+03
heavy_tailed	25%	500x500	auction_scaled	3	7.83	7.65	7.98
heavy_tailed	25%	500x500	csa	3	9.38	9.08	9.57
heavy_tailed	25%	500x500	hungarian	3	10.1	8.93	10.3
heavy_tailed	25%	500x500	auto	3	10.1	8.78	10.4
heavy_tailed	25%	500x500	jv	3	10.1	8.96	10.7
heavy_tailed	25%	500x500	sap	3	10.8	10.3	11.2
heavy_tailed	25%	500x500	csflow	3	11.2	10.2	11.4
heavy_tailed	25%	500x500	lapmod	3	12.8	10.8	13
heavy_tailed	25%	500x500	ramshaw_tarjan	3	21.4	18.5	21.5
heavy_tailed	25%	500x500	sap_dense	3	22.4	19.2	22.5
heavy_tailed	25%	500x500	network_simplex	3	98.6	93.8	100

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
heavy_tailed	25%	500x1500	auto	3	7.46	7.44	7.51
heavy_tailed	25%	500x1500	hungarian	3	7.5	7.45	7.51
heavy_tailed	25%	500x1500	jv	3	7.54	7.49	8.8
heavy_tailed	25%	500x1500	lapmod	3	8.32	8.31	8.33
heavy_tailed	25%	500x1500	ramshaw_tarjan	3	12.7	12.1	12.8
heavy_tailed	25%	500x1500	csflow	3	12.7	12.7	13.3
heavy_tailed	25%	500x1500	sap_dense	3	13.5	13.4	13.5
heavy_tailed	25%	500x1500	sap	3	14	13.4	14
heavy_tailed	25%	500x1500	csa	3	127	120	130
heavy_tailed	25%	500x1500	network_simplex	3	138	132	152
heavy_tailed	25%	500x1500	auction_scaled	3	235	229	251
heavy_tailed	25%	500x5000	hungarian	3	19.8	19.7	19.8
heavy_tailed	25%	500x5000	jv	3	23.7	21.7	23.8
heavy_tailed	25%	500x5000	auto	3	23.7	23.7	24.6
heavy_tailed	25%	500x5000	lapmod	3	26	24.7	26.6
heavy_tailed	25%	500x5000	ramshaw_tarjan	3	37.1	37	39.1
heavy_tailed	25%	500x5000	sap_dense	3	40.1	40.1	40.4
heavy_tailed	25%	500x5000	csflow	3	41.2	41	41.6
heavy_tailed	25%	500x5000	sap	3	46	43.8	46.2
heavy_tailed	25%	500x5000	network_simplex	3	691	686	748
heavy_tailed	25%	500x5000	csa	3	6.76e+03	6.67e+03	6.79e+03
heavy_tailed	60%	500x500	auto	3	8.87	8.83	9.44
heavy_tailed	60%	500x500	jv	3	8.87	8.81	9.33
heavy_tailed	60%	500x500	hungarian	3	9.58	9.16	9.81
heavy_tailed	60%	500x500	lapmod	3	13.3	12.9	14.1
heavy_tailed	60%	500x500	auction_scaled	3	14.4	14.4	14.5
heavy_tailed	60%	500x500	csa	3	14.5	14.3	14.7
heavy_tailed	60%	500x500	sap	3	16	16	16.8
heavy_tailed	60%	500x500	csflow	3	16.1	16	16.7
heavy_tailed	60%	500x500	sap_dense	3	19.7	19.6	21.3
heavy_tailed	60%	500x500	ramshaw_tarjan	3	20.9	20.8	22.5
heavy_tailed	60%	500x500	network_simplex	3	91.1	89.7	92.3
heavy_tailed	60%	500x1500	hungarian	3	8.49	8.46	8.58
heavy_tailed	60%	500x1500	jv	3	8.55	8.18	8.58
heavy_tailed	60%	500x1500	auto	3	8.56	8.17	8.6
heavy_tailed	60%	500x1500	lapmod	3	10.2	9.93	10.2
heavy_tailed	60%	500x1500	sap_dense	3	15.5	15.2	15.7
heavy_tailed	60%	500x1500	ramshaw_tarjan	3	17.6	17.5	17.8
heavy_tailed	60%	500x1500	sap	3	23	22.9	23.1
heavy_tailed	60%	500x1500	csflow	3	23.1	23	23.3
heavy_tailed	60%	500x1500	csa	3	133	129	139
heavy_tailed	60%	500x1500	network_simplex	3	173	165	191
heavy_tailed	60%	500x5000	jv	3	24.9	24.9	26.1
heavy_tailed	60%	500x5000	auto	3	25.5	25.2	26.4
heavy_tailed	60%	500x5000	hungarian	3	27	26.9	27.1
heavy_tailed	60%	500x5000	lapmod	3	33.2	32.4	33.2
heavy_tailed	60%	500x5000	sap_dense	3	47.6	47.4	47.9
heavy_tailed	60%	500x5000	ramshaw_tarjan	3	57.5	56.5	57.7
heavy_tailed	60%	500x5000	csflow	3	75.6	75.3	75.8
heavy_tailed	60%	500x5000	sap	3	76.1	75.4	77.3
heavy_tailed	60%	500x5000	network_simplex	3	740	726	751

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
heavy_tailed	60%	500x5000	csa	3	6.96e+03	6.92e+03	6.96e+03
int_uniform	4 blocks	500x500	csa	3	3.4	3.21	3.46
int_uniform	4 blocks	500x500	auction_scaled	3	3.61	3.23	3.69
int_uniform	4 blocks	500x500	auto	3	4.07	3.93	4.51
int_uniform	4 blocks	500x500	jv	3	4.11	4.1	4.55
int_uniform	4 blocks	500x500	hungarian	3	4.14	4.12	4.57
int_uniform	4 blocks	500x500	lapmod	3	5.06	4.91	5.31
int_uniform	4 blocks	500x500	csflow	3	5.39	5.29	5.46
int_uniform	4 blocks	500x500	sap	3	5.57	5.39	5.67
int_uniform	4 blocks	500x500	ramshaw_tarjan	3	5.75	5.62	6.15
int_uniform	4 blocks	500x500	sap_dense	3	6.96	6.72	7.1
int_uniform	4 blocks	500x500	network_simplex	3	30.7	29.8	31.5
int_uniform	4 blocks	500x1500	hungarian	3	3.77	3.7	4.84
int_uniform	4 blocks	500x1500	jv	3	3.78	3.73	4.26
int_uniform	4 blocks	500x1500	auto	3	3.8	3.73	4.28
int_uniform	4 blocks	500x1500	lapmod	3	4.41	4.3	5.45
int_uniform	4 blocks	500x1500	ramshaw_tarjan	3	7.67	7.55	8.14
int_uniform	4 blocks	500x1500	sap_dense	3	8.1	7.99	8.11
int_uniform	4 blocks	500x1500	sap	3	9.79	9.7	10.3
int_uniform	4 blocks	500x1500	csflow	3	10.7	10.1	10.7
int_uniform	4 blocks	500x1500	network_simplex	3	94.7	94.2	94.9
int_uniform	4 blocks	500x1500	csa	3	480	399	493
int_uniform	4 blocks	500x1500	auction_scaled	3	524	456	536
int_uniform	4 blocks	500x5000	jv	3	12	12	12.2
int_uniform	4 blocks	500x5000	hungarian	3	15.2	13.7	15.2
int_uniform	4 blocks	500x5000	auto	3	15.3	13.7	15.4
int_uniform	4 blocks	500x5000	lapmod	3	16.2	14.9	21.9
int_uniform	4 blocks	500x5000	ramshaw_tarjan	3	24.1	24	24.2
int_uniform	4 blocks	500x5000	sap_dense	3	29.2	29	30.6
int_uniform	4 blocks	500x5000	csflow	3	30.6	30.5	30.8
int_uniform	4 blocks	500x5000	sap	3	31.2	30.9	31.4
int_uniform	4 blocks	500x5000	network_simplex	3	548	547	561
int_uniform	4 blocks	500x5000	csa	3	4.9e+03	4.61e+03	4.95e+03
int_uniform	4 blocks	500x5000	auction_scaled	3	6.1e+03	6.06e+03	6.95e+03
int_uniform	complete	500x500	jv	3	4.62	4.25	4.66
int_uniform	complete	500x500	auto	3	4.64	4.25	4.7
int_uniform	complete	500x500	hungarian	3	8.66	8.6	8.89
int_uniform	complete	500x500	csa	3	10	10	10.2
int_uniform	complete	500x500	auction_scaled	3	10.1	9.93	10.2
int_uniform	complete	500x500	lapmod	3	12.5	12.3	12.8
int_uniform	complete	500x500	sap_dense	3	12.6	12.6	12.9
int_uniform	complete	500x500	ramshaw_tarjan	3	14.4	14.3	14.6
int_uniform	complete	500x500	csflow	3	20.9	20.9	21.9
int_uniform	complete	500x500	sap	3	21	20.8	21.3
int_uniform	complete	500x500	network_simplex	3	101	94.8	103
int_uniform	complete	500x1500	jv	3	4.72	4.71	4.76
int_uniform	complete	500x1500	auto	3	4.82	4.75	4.83
int_uniform	complete	500x1500	hungarian	3	4.84	4.77	4.86
int_uniform	complete	500x1500	lapmod	3	7.29	7.27	7.4
int_uniform	complete	500x1500	sap_dense	3	8.46	8.4	8.48
int_uniform	complete	500x1500	ramshaw_tarjan	3	16.2	15.7	16.4

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	complete	500x1500	sap	3	27.9	27.9	28.6
int_uniform	complete	500x1500	csflow	3	28.4	28.2	28.4
int_uniform	complete	500x1500	network_simplex	3	92	90.7	97.5
int_uniform	complete	500x1500	csa	3	390	384	427
int_uniform	complete	500x1500	auction_scaled	3	532	513	594
int_uniform	complete	500x5000	hungarian	3	13.8	13.8	13.8
int_uniform	complete	500x5000	jv	3	14	13.9	14
int_uniform	complete	500x5000	auto	3	14.1	14	14.1
int_uniform	complete	500x5000	lapmod	3	22.7	21	22.8
int_uniform	complete	500x5000	sap_dense	3	24.7	24.7	25.9
int_uniform	complete	500x5000	ramshaw_tarjan	3	48.6	48.6	49.9
int_uniform	complete	500x5000	csflow	3	103	103	104
int_uniform	complete	500x5000	sap	3	104	104	104
int_uniform	complete	500x5000	network_simplex	3	566	562	654
int_uniform	complete	500x5000	csa	3	4.72e+03	4.72e+03	4.85e+03
int_uniform	complete	500x5000	auction_scaled	3	4.76e+03	4.48e+03	5.27e+03
int_uniform	random_01	500x500	auction_scaled	3	1.42	1.36	1.59
int_uniform	random_01	500x500	csa	3	1.89	1.67	1.9
int_uniform	random_01	500x500	sap	3	2.88	2.73	2.88
int_uniform	random_01	500x500	csflow	3	4.01	3.4	4.21
int_uniform	random_01	500x500	jv	3	5.09	4.93	5.76
int_uniform	random_01	500x500	lapmod	3	5.5	5.09	5.92
int_uniform	random_01	500x500	auto	3	5.52	5.39	5.98
int_uniform	random_01	500x500	ramshaw_tarjan	3	5.88	5.72	6.38
int_uniform	random_01	500x500	hungarian	3	6.41	5.82	6.43
int_uniform	random_01	500x500	sap_dense	3	10.1	9.29	10.2
int_uniform	random_01	500x500	network_simplex	3	49.1	47.4	51.1
int_uniform	random_01	500x1500	jv	3	3.7	3.68	4.35
int_uniform	random_01	500x1500	auto	3	3.77	3.73	4.38
int_uniform	random_01	500x1500	csflow	3	3.83	3.82	4.45
int_uniform	random_01	500x1500	sap	3	4.01	3.99	4.61
int_uniform	random_01	500x1500	hungarian	3	4.93	4.32	4.96
int_uniform	random_01	500x1500	lapmod	3	5.31	4.7	5.34
int_uniform	random_01	500x1500	ramshaw_tarjan	3	7.02	6.33	7.02
int_uniform	random_01	500x1500	sap_dense	3	9.44	9.4	9.45
int_uniform	random_01	500x1500	network_simplex	3	65.5	62.4	76.8
int_uniform	random_01	500x1500	csa	3	452	442	469
int_uniform	random_01	500x1500	auction_scaled	3	508	508	620
int_uniform	random_01	500x5000	hungarian	3	12.2	12.2	19.1
int_uniform	random_01	500x5000	auto	3	12.4	12.4	14.7
int_uniform	random_01	500x5000	csflow	3	13.4	13.3	13.6
int_uniform	random_01	500x5000	jv	3	16.5	16.5	16.6
int_uniform	random_01	500x5000	lapmod	3	18	15.8	18
int_uniform	random_01	500x5000	sap	3	18.1	16	18.1
int_uniform	random_01	500x5000	ramshaw_tarjan	3	23.7	23.6	23.7
int_uniform	random_01	500x5000	sap_dense	3	27.6	25.9	27.8
int_uniform	random_01	500x5000	network_simplex	3	303	302	305
int_uniform	random_01	500x5000	csa	3	6.45e+03	5.83e+03	7.54e+03
int_uniform	random_01	500x5000	auction_scaled	3	8.21e+03	7.25e+03	9.05e+03
int_uniform	random_05	500x500	auction_scaled	3	2.34	2.32	2.34
int_uniform	random_05	500x500	csa	3	2.53	2.52	2.58

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	random_05	500x500	sap	3	5.13	4.78	5.18
int_uniform	random_05	500x500	csflow	3	5.44	5.13	5.57
int_uniform	random_05	500x500	jv	3	7.09	6.93	7.87
int_uniform	random_05	500x500	lapmod	3	7.14	7.02	7.54
int_uniform	random_05	500x500	hungarian	3	7.14	6.97	7.88
int_uniform	random_05	500x500	auto	3	7.15	6.96	7.44
int_uniform	random_05	500x500	ramshaw_tarjan	3	9.67	9.5	10.1
int_uniform	random_05	500x500	sap_dense	3	12.1	11.6	13.1
int_uniform	random_05	500x500	network_simplex	3	89.9	87.7	102
int_uniform	random_05	500x1500	csflow	3	5.27	5.26	5.89
int_uniform	random_05	500x1500	auto	3	5.36	4.76	5.36
int_uniform	random_05	500x1500	jv	3	5.36	4.76	5.37
int_uniform	random_05	500x1500	hungarian	3	5.39	5.36	5.4
int_uniform	random_05	500x1500	lapmod	3	5.85	5.84	5.87
int_uniform	random_05	500x1500	sap	3	6.62	6.62	6.62
int_uniform	random_05	500x1500	ramshaw_tarjan	3	8.01	7.92	9.32
int_uniform	random_05	500x1500	sap_dense	3	9.99	9.47	10
int_uniform	random_05	500x1500	network_simplex	3	109	102	116
int_uniform	random_05	500x1500	csa	3	449	429	462
int_uniform	random_05	500x1500	auction_scaled	3	464	462	468
int_uniform	random_05	500x5000	hungarian	3	13.5	13.4	14.6
int_uniform	random_05	500x5000	jv	3	13.6	13.5	15.8
int_uniform	random_05	500x5000	auto	3	13.7	13.6	15.7
int_uniform	random_05	500x5000	csflow	3	17.5	17.5	19.6
int_uniform	random_05	500x5000	lapmod	3	19.3	17.2	19.3
int_uniform	random_05	500x5000	sap	3	22.2	20.1	22.3
int_uniform	random_05	500x5000	ramshaw_tarjan	3	25.8	25.7	26
int_uniform	random_05	500x5000	sap_dense	3	35.9	34.2	36.3
int_uniform	random_05	500x5000	network_simplex	3	477	473	500
int_uniform	random_05	500x5000	csa	3	5.15e+03	5.03e+03	5.32e+03
int_uniform	random_05	500x5000	auction_scaled	3	6.12e+03	5.87e+03	8.21e+03
int_uniform	25%	500x500	auction_scaled	3	4.8	4.66	4.9
int_uniform	25%	500x500	csa	3	5.06	4.99	5.16
int_uniform	25%	500x500	jv	3	7.15	7.15	8.15
int_uniform	25%	500x500	hungarian	3	7.17	7.16	7.98
int_uniform	25%	500x500	auto	3	7.94	7.37	8.35
int_uniform	25%	500x500	lapmod	3	8.27	8.2	9.49
int_uniform	25%	500x500	csflow	3	9.24	9.08	9.95
int_uniform	25%	500x500	sap	3	9.36	9.3	10
int_uniform	25%	500x500	ramshaw_tarjan	3	15.1	14.5	16.7
int_uniform	25%	500x500	sap_dense	3	15.5	14.7	17.1
int_uniform	25%	500x500	network_simplex	3	106	102	108
int_uniform	25%	500x1500	auto	3	7.36	6.91	7.4
int_uniform	25%	500x1500	jv	3	7.37	6.86	7.41
int_uniform	25%	500x1500	hungarian	3	7.55	7.48	7.95
int_uniform	25%	500x1500	lapmod	3	8.23	7.72	8.28
int_uniform	25%	500x1500	csflow	3	12.6	12.4	13.2
int_uniform	25%	500x1500	ramshaw_tarjan	3	12.6	12.6	12.7
int_uniform	25%	500x1500	sap_dense	3	13.6	13.4	13.6
int_uniform	25%	500x1500	sap	3	13.9	13.8	14
int_uniform	25%	500x1500	network_simplex	3	106	105	134

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	25%	500x1500	auction_scaled	3	364	353	453
int_uniform	25%	500x1500	csa	3	464	388	496
int_uniform	25%	500x5000	hungarian	3	22.7	21.2	23
int_uniform	25%	500x5000	jv	3	23.6	21.6	23.6
int_uniform	25%	500x5000	auto	3	23.6	21.7	23.7
int_uniform	25%	500x5000	lapmod	3	27.1	25.2	27.2
int_uniform	25%	500x5000	ramshaw_tarjan	3	40.6	38.5	40.7
int_uniform	25%	500x5000	sap_dense	3	42.9	42.8	44.4
int_uniform	25%	500x5000	sap	3	43	42.4	44.6
int_uniform	25%	500x5000	csflow	3	45	43.1	47.4
int_uniform	25%	500x5000	network_simplex	3	630	626	643
int_uniform	25%	500x5000	auction_scaled	3	5.5e+03	5.42e+03	6.25e+03
int_uniform	25%	500x5000	csa	3	6.19e+03	5.57e+03	6.32e+03
int_uniform	60%	500x500	auction_scaled	3	8.57	8.45	8.88
int_uniform	60%	500x500	csa	3	8.73	8.67	8.74
int_uniform	60%	500x500	hungarian	3	9.06	9	9.09
int_uniform	60%	500x500	jv	3	9.09	8.95	9.38
int_uniform	60%	500x500	auto	3	9.53	9.42	9.65
int_uniform	60%	500x500	lapmod	3	12.9	12.5	13.1
int_uniform	60%	500x500	sap	3	16.5	16.4	16.5
int_uniform	60%	500x500	csflow	3	17	16.7	17.3
int_uniform	60%	500x500	sap_dense	3	20.4	20.3	20.5
int_uniform	60%	500x500	ramshaw_tarjan	3	21.8	21.5	22.4
int_uniform	60%	500x500	network_simplex	3	90.2	89.6	100
int_uniform	60%	500x1500	hungarian	3	8.68	8.64	8.77
int_uniform	60%	500x1500	auto	3	8.73	8.65	8.81
int_uniform	60%	500x1500	jv	3	8.75	8.34	8.8
int_uniform	60%	500x1500	lapmod	3	10.5	10	10.5
int_uniform	60%	500x1500	sap_dense	3	16.6	16.5	16.6
int_uniform	60%	500x1500	ramshaw_tarjan	3	18.3	18	18.5
int_uniform	60%	500x1500	csflow	3	23.4	23.2	23.9
int_uniform	60%	500x1500	sap	3	24.6	23.9	25.7
int_uniform	60%	500x1500	network_simplex	3	137	131	152
int_uniform	60%	500x1500	csa	3	369	359	385
int_uniform	60%	500x1500	auction_scaled	3	444	416	450
int_uniform	60%	500x5000	hungarian	3	25.5	25.4	26.3
int_uniform	60%	500x5000	jv	3	26.3	25.7	30.4
int_uniform	60%	500x5000	auto	3	27.2	26.8	27.2
int_uniform	60%	500x5000	lapmod	3	33.3	33.3	33.7
int_uniform	60%	500x5000	sap_dense	3	51.3	49.5	51.8
int_uniform	60%	500x5000	ramshaw_tarjan	3	58.3	57.1	59.1
int_uniform	60%	500x5000	csflow	3	77.6	77.1	77.6
int_uniform	60%	500x5000	sap	3	78.3	77.6	78.6
int_uniform	60%	500x5000	network_simplex	3	659	652	716
int_uniform	60%	500x5000	csa	3	4.53e+03	4.47e+03	4.72e+03
int_uniform	60%	500x5000	auction_scaled	3	5.39e+03	5.17e+03	5.61e+03
metric_clustered	4 blocks	500x500	auction_scaled	3	3.02	2.86	3.52
metric_clustered	4 blocks	500x500	csa	3	3.72	3.55	3.95
metric_clustered	4 blocks	500x500	jv	3	5.43	4.98	5.45
metric_clustered	4 blocks	500x500	hungarian	3	5.45	4.99	5.9
metric_clustered	4 blocks	500x500	auto	3	5.45	5.16	5.9

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_clustered	4 blocks	500x500	csflow	3	6.29	6.13	6.52
metric_clustered	4 blocks	500x500	lapmod	3	6.29	5.83	6.34
metric_clustered	4 blocks	500x500	sap	3	6.7	6.39	6.72
metric_clustered	4 blocks	500x500	ramshaw_tarjan	3	7.04	6.72	7.16
metric_clustered	4 blocks	500x500	sap_dense	3	8.8	8.25	8.82
metric_clustered	4 blocks	500x500	network_simplex	3	22.3	21.2	23
metric_clustered	4 blocks	500x1500	auto	3	3.92	3.89	3.95
metric_clustered	4 blocks	500x1500	jv	3	3.96	3.93	4.37
metric_clustered	4 blocks	500x1500	hungarian	3	3.96	3.94	4.38
metric_clustered	4 blocks	500x1500	lapmod	3	4.56	4.49	4.67
metric_clustered	4 blocks	500x1500	ramshaw_tarjan	3	7.3	7.23	7.38
metric_clustered	4 blocks	500x1500	sap_dense	3	9.83	9.37	9.86
metric_clustered	4 blocks	500x1500	csflow	3	10.3	9.73	10.4
metric_clustered	4 blocks	500x1500	sap	3	10.4	9.93	10.4
metric_clustered	4 blocks	500x1500	network_simplex	3	87.3	85.6	93.6
metric_clustered	4 blocks	500x1500	auction_scaled	3	356	351	370
metric_clustered	4 blocks	500x1500	csa	3	656	596	664
metric_clustered	4 blocks	500x5000	hungarian	3	12.1	12.1	12.2
metric_clustered	4 blocks	500x5000	jv	3	12.3	12.2	13.9
metric_clustered	4 blocks	500x5000	auto	3	13	12.6	14.2
metric_clustered	4 blocks	500x5000	lapmod	3	16.9	16.9	18.2
metric_clustered	4 blocks	500x5000	sap_dense	3	23.7	23.6	23.7
metric_clustered	4 blocks	500x5000	ramshaw_tarjan	3	24.5	24	25.1
metric_clustered	4 blocks	500x5000	csflow	3	30.2	30	30.5
metric_clustered	4 blocks	500x5000	sap	3	31.3	30.7	32.9
metric_clustered	4 blocks	500x5000	network_simplex	3	601	599	605
metric_clustered	4 blocks	500x5000	auction_scaled	3	4.6e+03	4.48e+03	4.63e+03
metric_clustered	4 blocks	500x5000	csa	3	1.34e+04	1.21e+04	1.4e+04
metric_clustered	complete	500x500	auto	3	6.33	6.13	7.48
metric_clustered	complete	500x500	jv	3	7.03	6.44	7.45
metric_clustered	complete	500x500	auction_scaled	3	7.69	7.49	8.29
metric_clustered	complete	500x500	csa	3	9.58	9.43	10.4
metric_clustered	complete	500x500	hungarian	3	9.95	9.51	11.4
metric_clustered	complete	500x500	sap_dense	3	14.3	13.8	16.5
metric_clustered	complete	500x500	lapmod	3	14.5	13.9	17.1
metric_clustered	complete	500x500	ramshaw_tarjan	3	17.6	17.3	19.2
metric_clustered	complete	500x500	sap	3	24.1	23.3	26.1
metric_clustered	complete	500x500	csflow	3	24.2	23.8	26.4
metric_clustered	complete	500x500	network_simplex	3	36.7	35.7	38.4
metric_clustered	complete	500x1500	auto	3	3.94	3.88	3.94
metric_clustered	complete	500x1500	jv	3	3.95	3.88	3.96
metric_clustered	complete	500x1500	hungarian	3	3.96	3.88	3.98
metric_clustered	complete	500x1500	lapmod	3	5.75	5.66	6.69
metric_clustered	complete	500x1500	sap_dense	3	8.28	8.18	9.44
metric_clustered	complete	500x1500	ramshaw_tarjan	3	13.5	13.5	14.8
metric_clustered	complete	500x1500	csflow	3	26.3	26.1	26.7
metric_clustered	complete	500x1500	sap	3	26.3	26.3	26.4
metric_clustered	complete	500x1500	network_simplex	3	102	102	107
metric_clustered	complete	500x1500	auction_scaled	3	350	341	379
metric_clustered	complete	500x1500	csa	3	513	500	612
metric_clustered	complete	500x5000	hungarian	3	11.2	11.1	15

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_clustered	complete	500x5000	jv	3	11.2	11.2	11.3
metric_clustered	complete	500x5000	auto	3	11.3	11.2	11.3
metric_clustered	complete	500x5000	lapmod	3	16.8	16.8	19.4
metric_clustered	complete	500x5000	sap_dense	3	21.3	21.3	23.6
metric_clustered	complete	500x5000	ramshaw_tarjan	3	45.9	45.6	46.6
metric_clustered	complete	500x5000	csflow	3	99	98.8	101
metric_clustered	complete	500x5000	sap	3	99.8	99.6	100
metric_clustered	complete	500x5000	network_simplex	3	609	600	637
metric_clustered	complete	500x5000	auction_scaled	3	4.09e+03	4.09e+03	4.53e+03
metric_clustered	complete	500x5000	csa	3	1.05e+04	1.04e+04	1.21e+04
metric_clustered	random_01	500x500	csa	3	1.55	1.55	1.78
metric_clustered	random_01	500x500	auction_scaled	3	1.74	1.53	1.77
metric_clustered	random_01	500x500	sap	3	2.68	2.63	2.96
metric_clustered	random_01	500x500	csflow	3	2.95	2.81	3.08
metric_clustered	random_01	500x500	jv	3	5.37	5.19	5.88
metric_clustered	random_01	500x500	auto	3	5.38	5.23	5.64
metric_clustered	random_01	500x500	lapmod	3	5.39	5.35	5.62
metric_clustered	random_01	500x500	hungarian	3	5.45	5.43	5.92
metric_clustered	random_01	500x500	ramshaw_tarjan	3	5.85	5.71	6.43
metric_clustered	random_01	500x500	sap_dense	3	8.94	8.63	9.28
metric_clustered	random_01	500x500	network_simplex	3	52.1	47.3	54.6
metric_clustered	random_01	500x1500	hungarian	3	3.74	3.73	4.45
metric_clustered	random_01	500x1500	auto	3	3.75	3.75	4.43
metric_clustered	random_01	500x1500	csflow	3	3.88	3.85	3.9
metric_clustered	random_01	500x1500	jv	3	5.02	5.01	5.11
metric_clustered	random_01	500x1500	sap	3	5.26	4.6	5.26
metric_clustered	random_01	500x1500	lapmod	3	5.36	4.77	5.38
metric_clustered	random_01	500x1500	ramshaw_tarjan	3	5.8	5.73	6.36
metric_clustered	random_01	500x1500	sap_dense	3	10.1	10	10.1
metric_clustered	random_01	500x1500	network_simplex	3	89.3	89.1	139
metric_clustered	random_01	500x1500	auction_scaled	3	491	446	509
metric_clustered	random_01	500x1500	csa	3	554	482	641
metric_clustered	random_01	500x5000	csflow	3	13.8	13.6	13.8
metric_clustered	random_01	500x5000	hungarian	3	16.5	16.5	16.5
metric_clustered	random_01	500x5000	jv	3	16.8	14.6	18
metric_clustered	random_01	500x5000	auto	3	16.8	16.7	16.8
metric_clustered	random_01	500x5000	lapmod	3	17.9	17.9	18
metric_clustered	random_01	500x5000	sap	3	18.1	16.2	22
metric_clustered	random_01	500x5000	ramshaw_tarjan	3	23.7	21.4	23.7
metric_clustered	random_01	500x5000	sap_dense	3	29.4	27	32.2
metric_clustered	random_01	500x5000	network_simplex	3	319	314	321
metric_clustered	random_01	500x5000	auction_scaled	3	5.69e+03	5.28e+03	5.75e+03
metric_clustered	random_01	500x5000	csa	3	1.06e+04	1.05e+04	1.17e+04
metric_clustered	random_05	500x500	auction_scaled	3	2.54	2.44	3.13
metric_clustered	random_05	500x500	csa	3	2.83	2.71	2.98
metric_clustered	random_05	500x500	sap	3	5.73	5.72	6.91
metric_clustered	random_05	500x500	csflow	3	6.18	5.9	6.69
metric_clustered	random_05	500x500	lapmod	3	8.56	7.99	10.1
metric_clustered	random_05	500x500	jv	3	9.03	8.74	10.4
metric_clustered	random_05	500x500	auto	3	9.05	8.31	10.2
metric_clustered	random_05	500x500	hungarian	3	9.07	8.44	10.4

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_clustered	random_05	500x500	ramshaw_tarjan	3	12.5	11.4	14.7
metric_clustered	random_05	500x500	sap_dense	3	15.1	13.9	17.5
metric_clustered	random_05	500x500	network_simplex	3	73.1	69.9	76.5
metric_clustered	random_05	500x1500	jv	3	5.52	4.92	5.53
metric_clustered	random_05	500x1500	hungarian	3	5.54	5.51	5.56
metric_clustered	random_05	500x1500	sap	3	5.55	5.5	6.16
metric_clustered	random_05	500x1500	auto	3	5.59	4.9	6.92
metric_clustered	random_05	500x1500	lapmod	3	5.99	5.96	5.99
metric_clustered	random_05	500x1500	csflow	3	6.55	6.54	7.02
metric_clustered	random_05	500x1500	ramshaw_tarjan	3	7.94	7.38	8
metric_clustered	random_05	500x1500	sap_dense	3	10.8	10.2	10.8
metric_clustered	random_05	500x1500	network_simplex	3	94.8	89.7	104
metric_clustered	random_05	500x1500	auction_scaled	3	405	370	468
metric_clustered	random_05	500x1500	csa	3	698	600	722
metric_clustered	random_05	500x5000	csflow	3	17.7	17.6	19.8
metric_clustered	random_05	500x5000	hungarian	3	17.8	17.8	17.8
metric_clustered	random_05	500x5000	jv	3	18	15.9	18
metric_clustered	random_05	500x5000	auto	3	18.1	18.1	18.1
metric_clustered	random_05	500x5000	lapmod	3	19.5	19.5	19.6
metric_clustered	random_05	500x5000	sap	3	22.6	20.4	24.5
metric_clustered	random_05	500x5000	ramshaw_tarjan	3	26	23.9	26.1
metric_clustered	random_05	500x5000	sap_dense	3	31.2	29.1	31.3
metric_clustered	random_05	500x5000	network_simplex	3	492	480	513
metric_clustered	random_05	500x5000	auction_scaled	3	4.63e+03	4.51e+03	5.09e+03
metric_clustered	random_05	500x5000	csa	3	1.05e+04	1.03e+04	1.11e+04
metric_clustered	25%	500x500	auction_scaled	3	4.95	4.94	4.96
metric_clustered	25%	500x500	csa	3	6.09	6.08	6.16
metric_clustered	25%	500x500	jv	3	13.6	13.1	14.1
metric_clustered	25%	500x500	auto	3	14	13.7	14.9
metric_clustered	25%	500x500	hungarian	3	14	13.3	15
metric_clustered	25%	500x500	csflow	3	14.9	14.3	15.2
metric_clustered	25%	500x500	sap	3	14.9	14.3	15.4
metric_clustered	25%	500x500	lapmod	3	16.7	16	17.4
metric_clustered	25%	500x500	ramshaw_tarjan	3	29.3	28.9	30.6
metric_clustered	25%	500x500	sap_dense	3	29.4	28.6	30.9
metric_clustered	25%	500x500	network_simplex	3	54.8	53.7	54.9
metric_clustered	25%	500x1500	jv	3	6.81	6.79	7.29
metric_clustered	25%	500x1500	auto	3	7.83	7.24	7.88
metric_clustered	25%	500x1500	hungarian	3	7.84	7.83	7.87
metric_clustered	25%	500x1500	lapmod	3	8.55	8.55	8.57
metric_clustered	25%	500x1500	sap_dense	3	13	12.8	13.7
metric_clustered	25%	500x1500	ramshaw_tarjan	3	13.3	13.2	13.3
metric_clustered	25%	500x1500	csflow	3	13.3	13.3	13.8
metric_clustered	25%	500x1500	sap	3	13.6	13.6	14.1
metric_clustered	25%	500x1500	network_simplex	3	117	113	118
metric_clustered	25%	500x1500	auction_scaled	3	341	329	358
metric_clustered	25%	500x1500	csa	3	588	536	601
metric_clustered	25%	500x5000	jv	3	20	19.9	22
metric_clustered	25%	500x5000	auto	3	22.7	21.3	23.4
metric_clustered	25%	500x5000	hungarian	3	23.1	21.7	23.4
metric_clustered	25%	500x5000	lapmod	3	27.3	27.3	27.4

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_clustered	25%	500x5000	ramshaw_tarjan	3	36.5	36.3	36.8
metric_clustered	25%	500x5000	sap_dense	3	37.2	37.2	37.4
metric_clustered	25%	500x5000	csflow	3	42.9	42.2	44.5
metric_clustered	25%	500x5000	sap	3	46.2	44.4	46.2
metric_clustered	25%	500x5000	network_simplex	3	663	659	664
metric_clustered	25%	500x5000	auction_scaled	3	4.13e+03	4.06e+03	4.95e+03
metric_clustered	25%	500x5000	csa	3	1.02e+04	1.02e+04	1.1e+04
metric_clustered	60%	500x500	auction_scaled	3	7.88	7.77	8.14
metric_clustered	60%	500x500	csa	3	9.84	9.74	9.84
metric_clustered	60%	500x500	jv	3	13.7	12.3	14.2
metric_clustered	60%	500x500	auto	3	13.8	12.4	14.3
metric_clustered	60%	500x500	hungarian	3	13.9	12.4	14.3
metric_clustered	60%	500x500	lapmod	3	21.4	18.3	21.7
metric_clustered	60%	500x500	sap	3	22.3	20.6	22.8
metric_clustered	60%	500x500	csflow	3	22.4	21.1	22.8
metric_clustered	60%	500x500	sap_dense	3	32	27.8	32.8
metric_clustered	60%	500x500	ramshaw_tarjan	3	36.6	32	37.5
metric_clustered	60%	500x500	network_simplex	3	39.5	38.4	39.6
metric_clustered	60%	500x1500	auto	3	9.07	8.81	9.17
metric_clustered	60%	500x1500	jv	3	9.11	8.72	9.19
metric_clustered	60%	500x1500	hungarian	3	9.13	9.05	9.19
metric_clustered	60%	500x1500	lapmod	3	10.7	10.7	10.8
metric_clustered	60%	500x1500	sap_dense	3	16.2	16.2	16.6
metric_clustered	60%	500x1500	ramshaw_tarjan	3	19.2	18.9	19.4
metric_clustered	60%	500x1500	csflow	3	24.4	24.3	24.6
metric_clustered	60%	500x1500	sap	3	24.6	24.3	25
metric_clustered	60%	500x1500	network_simplex	3	122	120	122
metric_clustered	60%	500x1500	auction_scaled	3	349	345	377
metric_clustered	60%	500x1500	csa	3	633	565	646
metric_clustered	60%	500x5000	jv	3	25.1	25	25.3
metric_clustered	60%	500x5000	hungarian	3	27.1	26.5	27.2
metric_clustered	60%	500x5000	auto	3	27.3	26.2	27.4
metric_clustered	60%	500x5000	lapmod	3	33.1	32.8	33.1
metric_clustered	60%	500x5000	sap_dense	3	47.4	47.4	47.9
metric_clustered	60%	500x5000	ramshaw_tarjan	3	58	58	58.1
metric_clustered	60%	500x5000	csflow	3	76.9	76.6	77
metric_clustered	60%	500x5000	sap	3	78.5	78	78.7
metric_clustered	60%	500x5000	network_simplex	3	660	656	663
metric_clustered	60%	500x5000	auction_scaled	3	4.25e+03	4.12e+03	4.3e+03
metric_clustered	60%	500x5000	csa	3	1.04e+04	1.02e+04	1.04e+04
metric_uniform	4 blocks	500x500	auction_scaled	3	2.49	2.47	2.6
metric_uniform	4 blocks	500x500	csa	3	3.56	3.5	3.58
metric_uniform	4 blocks	500x500	auto	3	4.52	4.37	4.52
metric_uniform	4 blocks	500x500	hungarian	3	4.54	4.53	4.69
metric_uniform	4 blocks	500x500	jv	3	4.55	4.55	4.78
metric_uniform	4 blocks	500x500	lapmod	3	5.22	5.21	5.4
metric_uniform	4 blocks	500x500	sap	3	5.66	5.65	5.75
metric_uniform	4 blocks	500x500	csflow	3	5.88	5.74	6
metric_uniform	4 blocks	500x500	ramshaw_tarjan	3	5.89	5.74	6.02
metric_uniform	4 blocks	500x500	sap_dense	3	7.46	7.33	7.53
metric_uniform	4 blocks	500x500	network_simplex	3	27.4	26.2	27.8

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_uniform	4 blocks	500x1500	auto	3	3.73	3.69	4.82
metric_uniform	4 blocks	500x1500	jv	3	3.75	3.72	4.17
metric_uniform	4 blocks	500x1500	lapmod	3	4.34	4.32	4.77
metric_uniform	4 blocks	500x1500	hungarian	3	4.62	4.16	4.66
metric_uniform	4 blocks	500x1500	ramshaw_tarjan	3	8.4	7.99	9.26
metric_uniform	4 blocks	500x1500	sap_dense	3	8.67	8.64	9.06
metric_uniform	4 blocks	500x1500	csflow	3	9.63	9.62	10.7
metric_uniform	4 blocks	500x1500	sap	3	9.87	9.86	10.2
metric_uniform	4 blocks	500x1500	network_simplex	3	93.9	90.1	94.6
metric_uniform	4 blocks	500x1500	auction_scaled	3	322	305	333
metric_uniform	4 blocks	500x1500	csa	3	597	512	603
metric_uniform	4 blocks	500x5000	hungarian	3	12.1	12.1	13.7
metric_uniform	4 blocks	500x5000	jv	3	12.1	12.1	13.8
metric_uniform	4 blocks	500x5000	auto	3	12.3	12.2	12.3
metric_uniform	4 blocks	500x5000	lapmod	3	16.1	14.9	17.8
metric_uniform	4 blocks	500x5000	sap_dense	3	23.5	23.4	25.2
metric_uniform	4 blocks	500x5000	ramshaw_tarjan	3	27.2	25.9	27.6
metric_uniform	4 blocks	500x5000	csflow	3	31.6	31.2	31.9
metric_uniform	4 blocks	500x5000	sap	3	34.2	33.7	34.5
metric_uniform	4 blocks	500x5000	network_simplex	3	511	510	514
metric_uniform	4 blocks	500x5000	auction_scaled	3	4.26e+03	4.14e+03	4.73e+03
metric_uniform	4 blocks	500x5000	csa	3	1.17e+04	1.11e+04	1.17e+04
metric_uniform	complete	500x500	auto	3	5.73	5.04	5.97
metric_uniform	complete	500x500	jv	3	5.78	5.04	5.98
metric_uniform	complete	500x500	auction_scaled	3	6.96	6.92	7.06
metric_uniform	complete	500x500	csa	3	9.09	9.03	9.3
metric_uniform	complete	500x500	hungarian	3	9.49	8.55	9.89
metric_uniform	complete	500x500	lapmod	3	13.7	12.1	14.5
metric_uniform	complete	500x500	sap_dense	3	13.8	12.5	14.5
metric_uniform	complete	500x500	ramshaw_tarjan	3	16	14.9	16.4
metric_uniform	complete	500x500	csflow	3	22.7	21.4	23.3
metric_uniform	complete	500x500	sap	3	22.7	21.5	23.2
metric_uniform	complete	500x500	network_simplex	3	70.2	68.1	82.9
metric_uniform	complete	500x1500	jv	3	3.72	3.7	3.86
metric_uniform	complete	500x1500	auto	3	3.76	3.72	3.88
metric_uniform	complete	500x1500	hungarian	3	3.76	3.72	3.97
metric_uniform	complete	500x1500	lapmod	3	5.36	5.27	5.42
metric_uniform	complete	500x1500	sap_dense	3	8.16	8.15	8.32
metric_uniform	complete	500x1500	ramshaw_tarjan	3	14.2	14.1	14.5
metric_uniform	complete	500x1500	sap	3	27.5	27.2	27.7
metric_uniform	complete	500x1500	csflow	3	28.2	27.8	28.3
metric_uniform	complete	500x1500	network_simplex	3	119	107	184
metric_uniform	complete	500x1500	auction_scaled	3	328	306	351
metric_uniform	complete	500x1500	csa	3	418	418	546
metric_uniform	complete	500x5000	hungarian	3	11.1	11	11.2
metric_uniform	complete	500x5000	jv	3	11.1	11.1	11.2
metric_uniform	complete	500x5000	auto	3	11.2	11.1	11.2
metric_uniform	complete	500x5000	sap_dense	3	21.2	21.1	21.2
metric_uniform	complete	500x5000	lapmod	3	21.8	19.1	25.7
metric_uniform	complete	500x5000	ramshaw_tarjan	3	48	46.8	48.1
metric_uniform	complete	500x5000	csflow	3	101	101	102

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_uniform	complete	500x5000	sap	3	101	101	104
metric_uniform	complete	500x5000	network_simplex	3	575	549	605
metric_uniform	complete	500x5000	auction_scaled	3	3.96e+03	3.95e+03	4.09e+03
metric_uniform	complete	500x5000	csa	3	1.18e+04	1.15e+04	1.19e+04
metric_uniform	random_01	500x500	auction_scaled	3	1.8	1.78	2.28
metric_uniform	random_01	500x500	csa	3	2.2	2.12	2.55
metric_uniform	random_01	500x500	sap	3	2.99	2.99	3.58
metric_uniform	random_01	500x500	csflow	3	3.44	3.2	3.65
metric_uniform	random_01	500x500	lapmod	3	5.4	5.13	5.59
metric_uniform	random_01	500x500	hungarian	3	5.49	5.45	6.03
metric_uniform	random_01	500x500	auto	3	5.5	5.44	5.57
metric_uniform	random_01	500x500	jv	3	5.55	5.5	5.6
metric_uniform	random_01	500x500	ramshaw_tarjan	3	5.97	5.88	6.06
metric_uniform	random_01	500x500	sap_dense	3	8.93	8.79	9.03
metric_uniform	random_01	500x500	network_simplex	3	60.4	55.3	64
metric_uniform	random_01	500x1500	hungarian	3	3.76	3.76	4.41
metric_uniform	random_01	500x1500	auto	3	3.8	3.78	4.41
metric_uniform	random_01	500x1500	csflow	3	3.85	3.84	4.48
metric_uniform	random_01	500x1500	sap	3	3.95	3.94	4.63
metric_uniform	random_01	500x1500	lapmod	3	4.14	4.13	4.79
metric_uniform	random_01	500x1500	jv	3	5.08	5.06	6.44
metric_uniform	random_01	500x1500	ramshaw_tarjan	3	5.69	5.68	5.71
metric_uniform	random_01	500x1500	sap_dense	3	8.86	8.59	9.28
metric_uniform	random_01	500x1500	network_simplex	3	104	102	106
metric_uniform	random_01	500x1500	auction_scaled	3	338	329	345
metric_uniform	random_01	500x1500	csa	3	543	526	626
metric_uniform	random_01	500x5000	csflow	3	13.8	13.6	13.9
metric_uniform	random_01	500x5000	hungarian	3	16.5	16.4	16.6
metric_uniform	random_01	500x5000	jv	3	16.6	14.5	16.6
metric_uniform	random_01	500x5000	auto	3	16.6	16.6	18
metric_uniform	random_01	500x5000	lapmod	3	18	18	18
metric_uniform	random_01	500x5000	sap	3	18.1	16	18.2
metric_uniform	random_01	500x5000	ramshaw_tarjan	3	23.7	21.5	23.7
metric_uniform	random_01	500x5000	sap_dense	3	25	24.8	26.9
metric_uniform	random_01	500x5000	network_simplex	3	330	324	333
metric_uniform	random_01	500x5000	auction_scaled	3	4.5e+03	4.45e+03	4.79e+03
metric_uniform	random_01	500x5000	csa	3	1.13e+04	1.1e+04	1.17e+04
metric_uniform	random_05	500x500	auction_scaled	3	2.31	2.29	2.32
metric_uniform	random_05	500x500	csa	3	2.61	2.61	2.62
metric_uniform	random_05	500x500	sap	3	5.12	4.98	5.49
metric_uniform	random_05	500x500	csflow	3	5.45	4.94	5.66
metric_uniform	random_05	500x500	hungarian	3	7.73	6.48	8.42
metric_uniform	random_05	500x500	auto	3	8.12	6.67	8.59
metric_uniform	random_05	500x500	jv	3	8.13	6.65	8.4
metric_uniform	random_05	500x500	lapmod	3	8.35	6.77	8.75
metric_uniform	random_05	500x500	ramshaw_tarjan	3	11.2	9.44	11.9
metric_uniform	random_05	500x500	sap_dense	3	13.8	11.7	14.3
metric_uniform	random_05	500x500	network_simplex	3	97.1	96.2	98.7
metric_uniform	random_05	500x1500	csflow	3	5.38	5.38	5.97
metric_uniform	random_05	500x1500	jv	3	5.47	5.46	5.49
metric_uniform	random_05	500x1500	hungarian	3	5.48	4.87	6.83

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_uniform	random_05	500x1500	auto	3	5.49	4.88	6.86
metric_uniform	random_05	500x1500	lapmod	3	5.94	5.93	7.3
metric_uniform	random_05	500x1500	sap	3	6.75	6.09	6.76
metric_uniform	random_05	500x1500	ramshaw_tarjan	3	6.77	6.75	7.4
metric_uniform	random_05	500x1500	sap_dense	3	10.6	10	10.7
metric_uniform	random_05	500x1500	network_simplex	3	270	183	301
metric_uniform	random_05	500x1500	auction_scaled	3	289	288	298
metric_uniform	random_05	500x1500	csa	3	534	524	534
metric_uniform	random_05	500x5000	auto	3	13.8	13.7	15.8
metric_uniform	random_05	500x5000	hungarian	3	17.7	15.6	17.8
metric_uniform	random_05	500x5000	jv	3	17.8	15.7	18
metric_uniform	random_05	500x5000	lapmod	3	19.4	19.4	19.5
metric_uniform	random_05	500x5000	csflow	3	21.9	19.8	21.9
metric_uniform	random_05	500x5000	sap	3	22.3	22.3	22.4
metric_uniform	random_05	500x5000	ramshaw_tarjan	3	25.9	23.9	25.9
metric_uniform	random_05	500x5000	sap_dense	3	31	29.1	31.1
metric_uniform	random_05	500x5000	network_simplex	3	513	497	536
metric_uniform	random_05	500x5000	auction_scaled	3	4.06e+03	3.9e+03	4.36e+03
metric_uniform	random_05	500x5000	csa	3	1.07e+04	1.07e+04	1.11e+04
metric_uniform	25%	500x500	auction_scaled	3	4.11	4.06	4.23
metric_uniform	25%	500x500	csa	3	5.56	5.37	5.61
metric_uniform	25%	500x500	jv	3	11	10.5	11.6
metric_uniform	25%	500x500	auto	3	11.1	10.1	11.1
metric_uniform	25%	500x500	hungarian	3	11.1	9.92	11.2
metric_uniform	25%	500x500	csflow	3	11.8	11.3	12.1
metric_uniform	25%	500x500	sap	3	12.3	11.6	12.3
metric_uniform	25%	500x500	lapmod	3	13.3	11.8	13.8
metric_uniform	25%	500x500	sap_dense	3	23.2	20.9	23.9
metric_uniform	25%	500x500	ramshaw_tarjan	3	24.4	21	24.4
metric_uniform	25%	500x500	network_simplex	3	89.4	88.8	93.7
metric_uniform	25%	500x1500	hungarian	3	6.42	6.38	7.02
metric_uniform	25%	500x1500	jv	3	7.49	6.91	7.49
metric_uniform	25%	500x1500	auto	3	7.5	7.49	8.7
metric_uniform	25%	500x1500	lapmod	3	8.29	8.29	8.3
metric_uniform	25%	500x1500	csflow	3	12.8	12.7	13.3
metric_uniform	25%	500x1500	ramshaw_tarjan	3	12.8	12.8	14
metric_uniform	25%	500x1500	sap	3	13.1	13	13.6
metric_uniform	25%	500x1500	sap_dense	3	13.7	13.3	13.9
metric_uniform	25%	500x1500	network_simplex	3	140	130	171
metric_uniform	25%	500x1500	auction_scaled	3	297	297	301
metric_uniform	25%	500x1500	csa	3	674	608	698
metric_uniform	25%	500x5000	jv	3	20	20	21.9
metric_uniform	25%	500x5000	hungarian	3	23.8	23.8	27.8
metric_uniform	25%	500x5000	auto	3	23.8	21.9	23.9
metric_uniform	25%	500x5000	lapmod	3	27.3	25.3	28.8
metric_uniform	25%	500x5000	ramshaw_tarjan	3	37.2	37.1	39.1
metric_uniform	25%	500x5000	sap_dense	3	41	40.5	41
metric_uniform	25%	500x5000	csflow	3	42.6	41.9	43.9
metric_uniform	25%	500x5000	sap	3	45	43.3	45.4
metric_uniform	25%	500x5000	network_simplex	3	635	630	679
metric_uniform	25%	500x5000	auction_scaled	3	3.99e+03	3.89e+03	4.19e+03

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
metric_uniform	25%	500x5000	csa	3	1.2e+04	1.13e+04	1.2e+04
metric_uniform	60%	500x500	auction_scaled	3	6.82	6.77	7.02
metric_uniform	60%	500x500	csa	3	8.95	8.89	9.08
metric_uniform	60%	500x500	jv	3	10	9.56	10.4
metric_uniform	60%	500x500	auto	3	10.1	9.54	10.4
metric_uniform	60%	500x500	hungarian	3	10.2	9.61	10.5
metric_uniform	60%	500x500	lapmod	3	15.2	13.8	15.7
metric_uniform	60%	500x500	sap	3	17.6	17	18.2
metric_uniform	60%	500x500	csflow	3	18.7	17.6	18.8
metric_uniform	60%	500x500	sap_dense	3	22.9	21.4	23.9
metric_uniform	60%	500x500	ramshaw_tarjan	3	25.9	24.3	27.1
metric_uniform	60%	500x500	network_simplex	3	82	81.1	82.7
metric_uniform	60%	500x1500	auto	3	7.94	7.86	7.99
metric_uniform	60%	500x1500	jv	3	7.97	7.85	8.35
metric_uniform	60%	500x1500	hungarian	3	8.62	8.2	8.72
metric_uniform	60%	500x1500	lapmod	3	10.3	9.94	10.3
metric_uniform	60%	500x1500	sap_dense	3	15.9	15.9	16.1
metric_uniform	60%	500x1500	ramshaw_tarjan	3	17.9	17.7	19.3
metric_uniform	60%	500x1500	sap	3	23.7	23.7	23.9
metric_uniform	60%	500x1500	csflow	3	24	23.8	24.1
metric_uniform	60%	500x1500	network_simplex	3	222	186	254
metric_uniform	60%	500x1500	auction_scaled	3	279	273	301
metric_uniform	60%	500x1500	csa	3	531	513	544
metric_uniform	60%	500x5000	hungarian	3	26.8	25.7	30.8
metric_uniform	60%	500x5000	auto	3	27	25.9	27.2
metric_uniform	60%	500x5000	jv	3	27.1	26	27.3
metric_uniform	60%	500x5000	lapmod	3	33.3	33.2	33.5
metric_uniform	60%	500x5000	sap_dense	3	46.8	45.9	48.8
metric_uniform	60%	500x5000	ramshaw_tarjan	3	59.6	57.5	62.5
metric_uniform	60%	500x5000	csflow	3	76	75.3	76
metric_uniform	60%	500x5000	sap	3	76.5	76.1	77.2
metric_uniform	60%	500x5000	network_simplex	3	631	614	662
metric_uniform	60%	500x5000	auction_scaled	3	3.53e+03	3.5e+03	3.79e+03
metric_uniform	60%	500x5000	csa	3	1.04e+04	1.02e+04	1.08e+04
tied	4 blocks	500x500	auction_scaled	3	4.15	4.08	4.25
tied	4 blocks	500x500	csa	3	4.54	4.38	4.55
tied	4 blocks	500x500	network_simplex	3	6.13	6.08	6.28
tied	4 blocks	500x500	csflow	3	13.4	13.4	13.8
tied	4 blocks	500x500	sap	3	13.5	13.4	13.5
tied	4 blocks	500x500	ramshaw_tarjan	3	16.1	16	16.2
tied	4 blocks	500x500	hungarian	3	17.9	17.8	18
tied	4 blocks	500x500	auto	3	17.9	17.9	18
tied	4 blocks	500x500	jv	3	17.9	17.9	18
tied	4 blocks	500x500	lapmod	3	19	18.9	19
tied	4 blocks	500x500	sap_dense	3	26.4	26.4	26.5
tied	4 blocks	500x1500	csflow	3	31	30.9	31.4
tied	4 blocks	500x1500	sap	3	31.1	31	31.4
tied	4 blocks	500x1500	network_simplex	3	32.8	32.7	33.1
tied	4 blocks	500x1500	ramshaw_tarjan	3	43.5	43.5	44.1
tied	4 blocks	500x1500	jv	3	54.2	54.2	54.6
tied	4 blocks	500x1500	hungarian	3	55	54.5	55.1

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
tied	4 blocks	500x1500	auto	3	55	54.5	55.1
tied	4 blocks	500x1500	lapmod	3	58.2	57.9	58.3
tied	4 blocks	500x1500	sap_dense	3	78	77.9	78.6
tied	4 blocks	500x1500	csa	3	104	104	107
tied	4 blocks	500x1500	auction_scaled	3	167	166	170
tied	4 blocks	500x5000	csflow	3	91.9	91.9	92.1
tied	4 blocks	500x5000	sap	3	92	91.9	93.3
tied	4 blocks	500x5000	ramshaw_tarjan	3	139	139	143
tied	4 blocks	500x5000	hungarian	3	182	181	183
tied	4 blocks	500x5000	jv	3	184	182	185
tied	4 blocks	500x5000	auto	3	184	184	186
tied	4 blocks	500x5000	lapmod	3	196	196	197
tied	4 blocks	500x5000	network_simplex	3	217	216	220
tied	4 blocks	500x5000	sap_dense	3	261	260	262
tied	4 blocks	500x5000	csa	3	1.69e+03	1.69e+03	1.7e+03
tied	4 blocks	500x5000	auction_scaled	3	2.66e+03	2.59e+03	2.74e+03
tied	complete	500x500	network_simplex	3	6.53	6.17	6.63
tied	complete	500x500	auction_scaled	3	13.8	13.6	14.5
tied	complete	500x500	csa	3	13.8	13.6	14.5
tied	complete	500x500	jv	3	57.5	57.2	57.7
tied	complete	500x500	auto	3	57.9	57.8	58.6
tied	complete	500x500	hungarian	3	61.8	61.6	61.9
tied	complete	500x500	ramshaw_tarjan	3	69.8	69.8	69.9
tied	complete	500x500	sap_dense	3	92.9	92.5	93.1
tied	complete	500x500	lapmod	3	97.5	97.3	97.6
tied	complete	500x500	sap	3	115	115	115
tied	complete	500x500	csflow	3	115	115	116
tied	complete	500x1500	network_simplex	3	23.4	23.3	23.9
tied	complete	500x1500	csa	3	94.9	94.1	108
tied	complete	500x1500	auction_scaled	3	150	146	152
tied	complete	500x1500	jv	3	203	203	204
tied	complete	500x1500	hungarian	3	203	203	204
tied	complete	500x1500	auto	3	204	203	204
tied	complete	500x1500	ramshaw_tarjan	3	226	226	226
tied	complete	500x1500	sap	3	275	274	275
tied	complete	500x1500	csflow	3	278	278	279
tied	complete	500x1500	sap_dense	3	290	289	290
tied	complete	500x1500	lapmod	3	298	298	299
tied	complete	500x5000	network_simplex	3	132	129	135
tied	complete	500x5000	hungarian	3	699	698	700
tied	complete	500x5000	jv	3	700	700	701
tied	complete	500x5000	auto	3	700	700	701
tied	complete	500x5000	ramshaw_tarjan	3	842	817	887
tied	complete	500x5000	sap	3	870	869	882
tied	complete	500x5000	csflow	3	879	879	908
tied	complete	500x5000	sap_dense	3	982	981	985
tied	complete	500x5000	lapmod	3	1.02e+03	1.01e+03	1.02e+03
tied	complete	500x5000	csa	3	1.45e+03	1.37e+03	1.46e+03
tied	complete	500x5000	auction_scaled	3	1.86e+03	1.84e+03	1.9e+03
tied	random_01	500x500	auction_scaled	3	1.85	1.57	1.94
tied	random_01	500x500	csa	3	2.02	1.74	2.02

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
tied	random_01	500x500	csflow	3	2.82	2.77	3.1
tied	random_01	500x500	sap	3	2.85	2.57	2.94
tied	random_01	500x500	lapmod	3	4.65	4.2	4.65
tied	random_01	500x500	hungarian	3	4.69	4.26	4.7
tied	random_01	500x500	auto	3	4.7	4.28	4.71
tied	random_01	500x500	jv	3	4.75	4.29	4.75
tied	random_01	500x500	ramshaw_tarjan	3	5.27	5.06	5.48
tied	random_01	500x500	sap_dense	3	7.85	7.26	7.92
tied	random_01	500x500	network_simplex	3	33.1	33	35.7
tied	random_01	500x1500	csflow	3	4.9	4.41	5.05
tied	random_01	500x1500	jv	3	5.23	4.6	5.24
tied	random_01	500x1500	auto	3	5.27	5.25	5.27
tied	random_01	500x1500	hungarian	3	5.29	5.24	5.29
tied	random_01	500x1500	sap	3	5.3	4.67	5.31
tied	random_01	500x1500	lapmod	3	5.53	4.94	5.56
tied	random_01	500x1500	ramshaw_tarjan	3	7.16	6.47	7.17
tied	random_01	500x1500	sap_dense	3	9.52	8.93	9.54
tied	random_01	500x1500	network_simplex	3	39.9	39.6	41.4
tied	random_01	500x1500	csa	3	241	224	241
tied	random_01	500x1500	auction_scaled	3	289	287	301
tied	random_01	500x5000	jv	3	13.7	13.5	15.7
tied	random_01	500x5000	hungarian	3	14	13.6	15.7
tied	random_01	500x5000	auto	3	17.9	15.6	18
tied	random_01	500x5000	csflow	3	18.3	16.1	18.3
tied	random_01	500x5000	lapmod	3	18.6	16.8	18.7
tied	random_01	500x5000	sap	3	18.7	16.4	18.8
tied	random_01	500x5000	ramshaw_tarjan	3	24.5	24.4	24.6
tied	random_01	500x5000	sap_dense	3	26.5	26.1	29.1
tied	random_01	500x5000	network_simplex	3	228	227	233
tied	random_01	500x5000	csa	3	1.71e+03	1.71e+03	1.8e+03
tied	random_01	500x5000	auction_scaled	3	2.75e+03	2.67e+03	2.77e+03
tied	random_05	500x500	auction_scaled	3	4.19	4.11	4.33
tied	random_05	500x500	csa	3	4.53	4.4	4.66
tied	random_05	500x500	csflow	3	21.3	21	21.6
tied	random_05	500x500	sap	3	21.7	21.6	21.8
tied	random_05	500x500	network_simplex	3	29.1	28.5	30.3
tied	random_05	500x500	auto	3	41.5	41.3	42.2
tied	random_05	500x500	hungarian	3	41.6	41.4	42.2
tied	random_05	500x500	jv	3	41.6	41.1	42.2
tied	random_05	500x500	lapmod	3	46.5	46.1	47.2
tied	random_05	500x500	ramshaw_tarjan	3	51.2	51	51.9
tied	random_05	500x500	sap_dense	3	70.4	70.3	71.3
tied	random_05	500x1500	csflow	3	39.6	39.5	41.6
tied	random_05	500x1500	network_simplex	3	40.4	39.7	40.5
tied	random_05	500x1500	sap	3	42.2	41.6	43.4
tied	random_05	500x1500	csa	3	102	98.1	107
tied	random_05	500x1500	auto	3	115	114	119
tied	random_05	500x1500	jv	3	115	113	119
tied	random_05	500x1500	hungarian	3	115	113	119
tied	random_05	500x1500	lapmod	3	118	117	122
tied	random_05	500x1500	ramshaw_tarjan	3	125	123	129

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
tied	random_05	500x1500	auction_scaled	3	149	147	149
tied	random_05	500x1500	sap_dense	3	189	188	196
tied	random_05	500x5000	csflow	3	113	110	117
tied	random_05	500x5000	sap	3	117	116	118
tied	random_05	500x5000	network_simplex	3	282	277	288
tied	random_05	500x5000	lapmod	3	385	371	388
tied	random_05	500x5000	auto	3	386	374	389
tied	random_05	500x5000	hungarian	3	390	380	391
tied	random_05	500x5000	jv	3	391	378	394
tied	random_05	500x5000	ramshaw_tarjan	3	395	380	396
tied	random_05	500x5000	sap_dense	3	628	611	635
tied	random_05	500x5000	csa	3	1.64e+03	1.63e+03	1.65e+03
tied	random_05	500x5000	auction_scaled	3	2.35e+03	2.19e+03	2.45e+03
tied	25%	500x500	auction_scaled	3	6.39	6.16	6.84
tied	25%	500x500	csa	3	6.54	6.43	6.57
tied	25%	500x500	network_simplex	3	9.71	9.46	9.82
tied	25%	500x500	sap	3	52.3	52.3	52.5
tied	25%	500x500	csflow	3	52.5	52.5	52.7
tied	25%	500x500	hungarian	3	63.4	62.7	63.4
tied	25%	500x500	jv	3	63.6	63.2	63.6
tied	25%	500x500	auto	3	63.6	62.9	63.6
tied	25%	500x500	lapmod	3	74.1	73.9	74.1
tied	25%	500x500	ramshaw_tarjan	3	109	109	109
tied	25%	500x500	sap_dense	3	128	128	128
tied	25%	500x1500	network_simplex	3	37.2	36.7	37.4
tied	25%	500x1500	csa	3	98.8	98	100
tied	25%	500x1500	sap	3	116	116	116
tied	25%	500x1500	csflow	3	117	117	117
tied	25%	500x1500	auction_scaled	3	139	129	141
tied	25%	500x1500	hungarian	3	203	203	203
tied	25%	500x1500	auto	3	203	203	203
tied	25%	500x1500	jv	3	204	204	204
tied	25%	500x1500	lapmod	3	223	223	223
tied	25%	500x1500	ramshaw_tarjan	3	403	402	403
tied	25%	500x1500	sap_dense	3	473	472	474
tied	25%	500x5000	network_simplex	3	236	232	243
tied	25%	500x5000	sap	3	349	346	356
tied	25%	500x5000	csflow	3	351	349	354
tied	25%	500x5000	jv	3	704	701	706
tied	25%	500x5000	auto	3	704	702	706
tied	25%	500x5000	hungarian	3	704	702	706
tied	25%	500x5000	lapmod	3	745	745	747
tied	25%	500x5000	csa	3	1.35e+03	1.26e+03	1.35e+03
tied	25%	500x5000	ramshaw_tarjan	3	1.42e+03	1.42e+03	1.43e+03
tied	25%	500x5000	sap_dense	3	1.68e+03	1.68e+03	1.69e+03
tied	25%	500x5000	auction_scaled	3	1.75e+03	1.72e+03	1.81e+03
tied	60%	500x500	network_simplex	3	8.17	8.15	8.42
tied	60%	500x500	auction_scaled	3	11.5	10.9	11.6
tied	60%	500x500	csa	3	12.1	11.3	12.3
tied	60%	500x500	auto	3	63.9	63.5	64.1
tied	60%	500x500	hungarian	3	64.5	64.3	64.6

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
tied	60%	500x500	jv	3	64.8	64.1	65
tied	60%	500x500	sap	3	83.3	83.1	83.4
tied	60%	500x500	csflow	3	83.8	83.7	83.8
tied	60%	500x500	lapmod	3	85.2	85.1	85.5
tied	60%	500x500	sap_dense	3	134	133	134
tied	60%	500x500	ramshaw_tarjan	3	146	146	146
tied	60%	500x1500	network_simplex	3	38.6	37.2	39.5
tied	60%	500x1500	csa	3	106	103	108
tied	60%	500x1500	auction_scaled	3	142	137	147
tied	60%	500x1500	sap	3	195	195	196
tied	60%	500x1500	csflow	3	198	197	198
tied	60%	500x1500	jv	3	208	208	210
tied	60%	500x1500	hungarian	3	209	208	209
tied	60%	500x1500	auto	3	209	209	209
tied	60%	500x1500	lapmod	3	262	262	262
tied	60%	500x1500	ramshaw_tarjan	3	552	551	552
tied	60%	500x1500	sap_dense	3	584	584	585
tied	60%	500x5000	network_simplex	3	197	196	197
tied	60%	500x5000	sap	3	617	615	630
tied	60%	500x5000	csflow	3	633	626	642
tied	60%	500x5000	jv	3	715	714	716
tied	60%	500x5000	auto	3	716	715	716
tied	60%	500x5000	hungarian	3	716	715	716
tied	60%	500x5000	lapmod	3	885	884	885
tied	60%	500x5000	csa	3	1.38e+03	1.29e+03	1.46e+03
tied	60%	500x5000	auction_scaled	3	1.52e+03	1.51e+03	1.7e+03
tied	60%	500x5000	ramshaw_tarjan	3	2.08e+03	2.02e+03	2.1e+03
tied	60%	500x5000	sap_dense	3	2.15e+03	2.15e+03	2.15e+03
binary	complete	1000x10000	auto	2	174	173	175
binary	complete	1000x10000	jv	2	5.43e+03	5.43e+03	5.43e+03
binary	complete	1000x10000	sap	2	5.99e+03	5.99e+03	5.99e+03
binary	complete	1000x10000	ramshaw_tarjan	2	6.32e+03	6.32e+03	6.32e+03
binary	complete	1000x10000	lapmod	2	7.95e+03	7.94e+03	7.95e+03
binary	complete	1500x1500	auto	2	35	34.8	35.1
binary	complete	1500x1500	ramshaw_tarjan	2	1.5e+03	1.5e+03	1.5e+03
binary	complete	1500x1500	jv	2	1.51e+03	1.51e+03	1.51e+03
binary	complete	1500x1500	lapmod	2	2.51e+03	2.51e+03	2.51e+03
binary	complete	1500x1500	sap	2	2.51e+03	2.51e+03	2.52e+03
binary	complete	1500x4500	auto	2	119	118	120
binary	complete	1500x4500	jv	2	5.24e+03	5.24e+03	5.24e+03
binary	complete	1500x4500	ramshaw_tarjan	2	5.49e+03	5.49e+03	5.5e+03
binary	complete	1500x4500	sap	2	6.43e+03	6.39e+03	6.47e+03
binary	complete	1500x4500	lapmod	2	7.86e+03	7.85e+03	7.86e+03
binary	random_01	1000x10000	auto	2	82.7	82.2	83.3
binary	random_01	1000x10000	sap	2	320	320	321
binary	random_01	1000x10000	ramshaw_tarjan	2	2.02e+03	2.02e+03	2.02e+03
binary	random_01	1000x10000	lapmod	2	2.89e+03	2.88e+03	2.9e+03
binary	random_01	1000x10000	jv	2	3.06e+03	3.06e+03	3.07e+03
binary	random_01	1500x1500	auto	2	14.9	14	15.8
binary	random_01	1500x1500	sap	2	193	193	194
binary	random_01	1500x1500	ramshaw_tarjan	2	945	941	948

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
binary	random_01	1500x1500	jv	2	1.26e+03	1.25e+03	1.27e+03
binary	random_01	1500x1500	lapmod	2	1.4e+03	1.4e+03	1.41e+03
binary	random_01	1500x4500	auto	2	55.3	55.1	55.4
binary	random_01	1500x4500	sap	2	381	380	383
binary	random_01	1500x4500	ramshaw_tarjan	2	2.7e+03	2.69e+03	2.71e+03
binary	random_01	1500x4500	lapmod	2	4.09e+03	4.08e+03	4.11e+03
binary	random_01	1500x4500	jv	2	4.19e+03	4.17e+03	4.21e+03
binary	25%	1000x10000	auto	2	137	137	137
binary	25%	1000x10000	sap	2	2.06e+03	2.06e+03	2.06e+03
binary	25%	1000x10000	jv	2	5.5e+03	5.5e+03	5.5e+03
binary	25%	1000x10000	lapmod	2	5.96e+03	5.95e+03	5.96e+03
binary	25%	1000x10000	ramshaw_tarjan	2	1.05e+04	1.05e+04	1.05e+04
binary	25%	1500x1500	auto	2	26.8	26.7	26.8
binary	25%	1500x1500	sap	2	815	814	816
binary	25%	1500x1500	jv	2	1.54e+03	1.54e+03	1.54e+03
binary	25%	1500x1500	lapmod	2	1.89e+03	1.89e+03	1.89e+03
binary	25%	1500x1500	ramshaw_tarjan	2	2.69e+03	2.69e+03	2.69e+03
binary	25%	1500x4500	auto	2	93.4	93.1	93.8
binary	25%	1500x4500	sap	2	2e+03	2e+03	2e+03
binary	25%	1500x4500	jv	2	5.29e+03	5.29e+03	5.29e+03
binary	25%	1500x4500	lapmod	2	5.78e+03	5.78e+03	5.78e+03
binary	25%	1500x4500	ramshaw_tarjan	2	1e+04	1e+04	1.01e+04
dbl_uniform	complete	1000x10000	auto	2	79.6	79.3	79.9
dbl_uniform	complete	1000x10000	jv	2	80.1	80	80.1
dbl_uniform	complete	1000x10000	lapmod	2	101	101	101
dbl_uniform	complete	1000x10000	ramshaw_tarjan	2	229	228	230
dbl_uniform	complete	1000x10000	sap	2	480	479	481
dbl_uniform	complete	1500x1500	auto	2	48.8	48.7	48.9
dbl_uniform	complete	1500x1500	jv	2	48.8	48.7	48.9
dbl_uniform	complete	1500x1500	ramshaw_tarjan	2	156	155	157
dbl_uniform	complete	1500x1500	lapmod	2	165	164	167
dbl_uniform	complete	1500x1500	sap	2	256	254	259
dbl_uniform	complete	1500x4500	jv	2	59.9	59.9	59.9
dbl_uniform	complete	1500x4500	auto	2	60.2	60.2	60.2
dbl_uniform	complete	1500x4500	lapmod	2	74.8	74.1	75.5
dbl_uniform	complete	1500x4500	ramshaw_tarjan	2	162	162	163
dbl_uniform	complete	1500x4500	sap	2	323	323	324
dbl_uniform	random_01	1000x10000	auto	2	83.5	83.4	83.5
dbl_uniform	random_01	1000x10000	jv	2	86.8	84.8	88.9
dbl_uniform	random_01	1000x10000	lapmod	2	92.6	90.6	94.7
dbl_uniform	random_01	1000x10000	sap	2	95.3	95.2	95.4
dbl_uniform	random_01	1000x10000	ramshaw_tarjan	2	121	121	121
dbl_uniform	random_01	1500x1500	sap	2	32.2	31.8	32.5
dbl_uniform	random_01	1500x1500	jv	2	80.9	78.3	83.5
dbl_uniform	random_01	1500x1500	ramshaw_tarjan	2	83.5	81.1	86
dbl_uniform	random_01	1500x1500	lapmod	2	83.7	81.4	86.1
dbl_uniform	random_01	1500x1500	auto	2	84.3	81.7	86.9
dbl_uniform	random_01	1500x4500	jv	2	62.9	60.4	65.5
dbl_uniform	random_01	1500x4500	lapmod	2	66	63.4	68.6
dbl_uniform	random_01	1500x4500	auto	2	69.1	68.9	69.3
dbl_uniform	random_01	1500x4500	sap	2	72.6	71.4	73.8

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
dbl_uniform	random_01	1500x4500	ramshaw_tarjan	2	92.5	86	99
dbl_uniform	25%	1000x10000	auto	2	113	113	113
dbl_uniform	25%	1000x10000	jv	2	116	115	118
dbl_uniform	25%	1000x10000	lapmod	2	124	124	124
dbl_uniform	25%	1000x10000	ramshaw_tarjan	2	195	192	199
dbl_uniform	25%	1000x10000	sap	2	221	220	223
dbl_uniform	25%	1500x1500	jv	2	97.8	92.2	103
dbl_uniform	25%	1500x1500	auto	2	99.3	94.5	104
dbl_uniform	25%	1500x1500	sap	2	106	103	109
dbl_uniform	25%	1500x1500	lapmod	2	123	115	131
dbl_uniform	25%	1500x1500	ramshaw_tarjan	2	213	200	227
dbl_uniform	25%	1500x4500	jv	2	86.1	84.1	88.2
dbl_uniform	25%	1500x4500	auto	2	87.3	83.7	90.8
dbl_uniform	25%	1500x4500	lapmod	2	93	91.6	94.3
dbl_uniform	25%	1500x4500	ramshaw_tarjan	2	140	140	140
dbl_uniform	25%	1500x4500	sap	2	154	152	157
heavy_tailed	complete	1000x10000	jv	2	77.9	77.9	78
heavy_tailed	complete	1000x10000	auto	2	78.2	78.2	78.2
heavy_tailed	complete	1000x10000	lapmod	2	98.9	98.6	99.2
heavy_tailed	complete	1000x10000	ramshaw_tarjan	2	231	228	233
heavy_tailed	complete	1000x10000	sap	2	482	480	483
heavy_tailed	complete	1500x1500	jv	2	47.1	46	48.2
heavy_tailed	complete	1500x1500	auto	2	47.2	46.1	48.3
heavy_tailed	complete	1500x1500	ramshaw_tarjan	2	144	144	145
heavy_tailed	complete	1500x1500	lapmod	2	145	144	146
heavy_tailed	complete	1500x1500	sap	2	240	239	242
heavy_tailed	complete	1500x4500	auto	2	60.6	60.3	60.9
heavy_tailed	complete	1500x4500	jv	2	61.2	61.1	61.3
heavy_tailed	complete	1500x4500	lapmod	2	74.6	74.4	74.8
heavy_tailed	complete	1500x4500	ramshaw_tarjan	2	163	163	163
heavy_tailed	complete	1500x4500	sap	2	328	328	329
heavy_tailed	random_01	1000x10000	auto	2	82.8	82.7	82.9
heavy_tailed	random_01	1000x10000	jv	2	85.5	84.1	87
heavy_tailed	random_01	1000x10000	lapmod	2	88.6	88.5	88.6
heavy_tailed	random_01	1000x10000	sap	2	100	98.5	102
heavy_tailed	random_01	1000x10000	ramshaw_tarjan	2	127	124	129
heavy_tailed	random_01	1500x1500	sap	2	35.2	35.2	35.2
heavy_tailed	random_01	1500x1500	lapmod	2	110	103	117
heavy_tailed	random_01	1500x1500	ramshaw_tarjan	2	112	106	117
heavy_tailed	random_01	1500x1500	jv	2	112	106	118
heavy_tailed	random_01	1500x1500	auto	2	112	106	118
heavy_tailed	random_01	1500x4500	jv	2	61.4	59.4	63.3
heavy_tailed	random_01	1500x4500	lapmod	2	65.7	63	68.5
heavy_tailed	random_01	1500x4500	auto	2	67.9	67.8	68
heavy_tailed	random_01	1500x4500	sap	2	69.4	69.3	69.6
heavy_tailed	random_01	1500x4500	ramshaw_tarjan	2	90.2	90.1	90.3
heavy_tailed	25%	1000x10000	auto	2	112	112	112
heavy_tailed	25%	1000x10000	jv	2	117	115	119
heavy_tailed	25%	1000x10000	lapmod	2	130	130	130
heavy_tailed	25%	1000x10000	ramshaw_tarjan	2	189	188	189
heavy_tailed	25%	1000x10000	sap	2	222	220	224

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
heavy_tailed	25%	1500x1500	jv	2	99	97.7	100
heavy_tailed	25%	1500x1500	auto	2	101	99.1	104
heavy_tailed	25%	1500x1500	sap	2	106	105	107
heavy_tailed	25%	1500x1500	lapmod	2	122	122	123
heavy_tailed	25%	1500x1500	ramshaw_tarjan	2	213	212	214
heavy_tailed	25%	1500x4500	jv	2	85	82.6	87.4
heavy_tailed	25%	1500x4500	auto	2	86.1	83.2	89.1
heavy_tailed	25%	1500x4500	lapmod	2	95.1	95	95.2
heavy_tailed	25%	1500x4500	ramshaw_tarjan	2	137	135	138
heavy_tailed	25%	1500x4500	sap	2	159	158	159
int_uniform	complete	1000x10000	jv	2	85.2	85.1	85.3
int_uniform	complete	1000x10000	auto	2	85.5	85.5	85.6
int_uniform	complete	1000x10000	lapmod	2	106	106	107
int_uniform	complete	1000x10000	ramshaw_tarjan	2	238	235	240
int_uniform	complete	1000x10000	sap	2	489	489	489
int_uniform	complete	1500x1500	jv	2	45.1	44.6	45.6
int_uniform	complete	1500x1500	auto	2	45.3	44.8	45.8
int_uniform	complete	1500x1500	lapmod	2	133	131	135
int_uniform	complete	1500x1500	ramshaw_tarjan	2	139	137	141
int_uniform	complete	1500x1500	sap	2	228	226	231
int_uniform	complete	1500x4500	jv	2	64.3	64	64.5
int_uniform	complete	1500x4500	auto	2	64.9	64.8	65
int_uniform	complete	1500x4500	lapmod	2	78.3	78.1	78.4
int_uniform	complete	1500x4500	ramshaw_tarjan	2	169	168	170
int_uniform	complete	1500x4500	sap	2	329	328	330
int_uniform	random_01	1000x10000	jv	2	82	81.4	82.6
int_uniform	random_01	1000x10000	auto	2	86.4	84.6	88.1
int_uniform	random_01	1000x10000	lapmod	2	88.6	88.5	88.7
int_uniform	random_01	1000x10000	sap	2	94.7	94.2	95.2
int_uniform	random_01	1000x10000	ramshaw_tarjan	2	123	122	125
int_uniform	random_01	1500x1500	sap	2	30.9	29.8	31.9
int_uniform	random_01	1500x1500	ramshaw_tarjan	2	87.6	86.2	89
int_uniform	random_01	1500x1500	auto	2	89.2	88.4	90.1
int_uniform	random_01	1500x1500	jv	2	90.3	87.9	92.8
int_uniform	random_01	1500x1500	lapmod	2	91.5	90.7	92.2
int_uniform	random_01	1500x4500	auto	2	61.8	59.3	64.3
int_uniform	random_01	1500x4500	jv	2	63.6	62	65.1
int_uniform	random_01	1500x4500	lapmod	2	65.3	62.7	68
int_uniform	random_01	1500x4500	sap	2	69.6	67	72.2
int_uniform	random_01	1500x4500	ramshaw_tarjan	2	84.5	82	87
int_uniform	25%	1000x10000	auto	2	111	111	111
int_uniform	25%	1000x10000	jv	2	111	111	111
int_uniform	25%	1000x10000	lapmod	2	123	123	124
int_uniform	25%	1000x10000	ramshaw_tarjan	2	185	185	186
int_uniform	25%	1000x10000	sap	2	220	218	222
int_uniform	25%	1500x1500	auto	2	97.1	96.5	97.8
int_uniform	25%	1500x1500	jv	2	97.2	96.4	97.9
int_uniform	25%	1500x1500	sap	2	107	106	108
int_uniform	25%	1500x1500	lapmod	2	125	122	127
int_uniform	25%	1500x1500	ramshaw_tarjan	2	218	215	221
int_uniform	25%	1500x4500	jv	2	82.6	80.2	84.9

Regime	Pattern	Shape	Solver	Instances	Median (ms)	Q25 (ms)	Q75 (ms)
int_uniform	25%	1500x4500	auto	2	86.7	85.9	87.4
int_uniform	25%	1500x4500	lapmod	2	94.3	94	94.6
int_uniform	25%	1500x4500	ramshaw_tarjan	2	135	133	136
int_uniform	25%	1500x4500	sap	2	153	152	155
metric_clustered	complete	1000x10000	jv	2	77.7	77.6	77.8
metric_clustered	complete	1000x10000	auto	2	78.3	78	78.5
metric_clustered	complete	1000x10000	lapmod	2	97.9	97.2	98.5
metric_clustered	complete	1000x10000	ramshaw_tarjan	2	229	225	232
metric_clustered	complete	1000x10000	sap	2	479	479	480
metric_clustered	complete	1500x1500	auto	2	154	141	167
metric_clustered	complete	1500x1500	jv	2	154	141	167
metric_clustered	complete	1500x1500	ramshaw_tarjan	2	271	256	286
metric_clustered	complete	1500x1500	lapmod	2	313	291	334
metric_clustered	complete	1500x1500	sap	2	415	394	436
metric_clustered	complete	1500x4500	auto	2	59.6	59.1	60.1
metric_clustered	complete	1500x4500	jv	2	59.8	59.1	60.4
metric_clustered	complete	1500x4500	lapmod	2	74.8	73.9	75.6
metric_clustered	complete	1500x4500	ramshaw_tarjan	2	163	162	164
metric_clustered	complete	1500x4500	sap	2	325	322	327
metric_clustered	random_01	1000x10000	jv	2	83.3	83.2	83.4
metric_clustered	random_01	1000x10000	auto	2	86.5	84.9	88.1
metric_clustered	random_01	1000x10000	lapmod	2	88.7	88.6	88.8
metric_clustered	random_01	1000x10000	sap	2	101	98.9	102
metric_clustered	random_01	1000x10000	ramshaw_tarjan	2	121	121	121
metric_clustered	random_01	1500x1500	sap	2	33.6	33.3	33.9
metric_clustered	random_01	1500x1500	jv	2	91.8	89.9	93.6
metric_clustered	random_01	1500x1500	auto	2	92.8	91.4	94.2
metric_clustered	random_01	1500x1500	ramshaw_tarjan	2	92.9	89.8	96
metric_clustered	random_01	1500x1500	lapmod	2	93.6	91.1	96
metric_clustered	random_01	1500x4500	lapmod	2	66	63.2	68.7
metric_clustered	random_01	1500x4500	auto	2	69.1	63.8	74.3
metric_clustered	random_01	1500x4500	jv	2	69.5	65.7	73.3
metric_clustered	random_01	1500x4500	sap	2	73.6	72.1	75.1
metric_clustered	random_01	1500x4500	ramshaw_tarjan	2	97.9	94.1	102
metric_clustered	25%	1000x10000	auto	2	113	113	113
metric_clustered	25%	1000x10000	jv	2	118	116	119
metric_clustered	25%	1000x10000	lapmod	2	123	123	123
metric_clustered	25%	1000x10000	ramshaw_tarjan	2	189	188	191
metric_clustered	25%	1000x10000	sap	2	223	223	224
metric_clustered	25%	1500x1500	sap	2	144	140	148
metric_clustered	25%	1500x1500	jv	2	151	144	157
metric_clustered	25%	1500x1500	auto	2	152	147	156
metric_clustered	25%	1500x1500	lapmod	2	186	178	194
metric_clustered	25%	1500x1500	ramshaw_tarjan	2	337	323	351
metric_clustered	25%	1500x4500	jv	2	84.2	83.9	84.4
metric_clustered	25%	1500x4500	auto	2	84.2	83.9	84.6
metric_clustered	25%	1500x4500	lapmod	2	89.6	88.5	90.7
metric_clustered	25%	1500x4500	ramshaw_tarjan	2	141	139	142
metric_clustered	25%	1500x4500	sap	2	164	160	168

4.4 Do the solvers agree

Every solver in a cell solves the same instance, so their totals have to agree. A disagreement would be a defect in a solver rather than a property of the regime.

522 instances carrying 5 to 12 solvers each; the worst relative disagreement on the optimum is 696.

5 The edge-generation loop away from one cloud

The article’s implicit results come from one cloud: eight independent normal covariates with the treated group shifted, one draw per size. This grid sweeps one factor at a time around that base, several seeds deep. Where the dense path can afford the problem, the loop’s answer is compared against it unit by unit.

Table 7: The loop by cell. Rounds is the range over seeds. Graph is the median share of the complete pair count the candidate set ended up holding, in percent. Distances is the median number of distance evaluations over all rounds as a multiple of the complete pair count. Implicit and lazy are median seconds over seeds and speedup is their median ratio.

Sweep	Cloud	Dim	Units	Caliper	Seeds	Rounds	Graph %	Distances	Implicit	Lazy	Speedup
caliper	gaussian	8	10,000	boundary	5	2-2	1.151	0.31x	0.91 s	1.66 s	1.79x
caliper	gaussian	8	10,000	mid	5	2-2	1.163	0.51x	0.989 s	1.8 s	1.82x
caliper	gaussian	8	10,000	none	5	2-2	1.170	1.79x	0.863 s	1.77 s	2.05x
cloud	clustered	8	10,000	none	5	2-2	1.170	0.48x	0.324 s	0.654 s	2.02x
cloud	contested	8	10,000	none	5	9-10	73.318	6.67x	134 s	426 s	3.17x
cloud	gaussian	8	10,000	none	5	2-2	1.170	1.79x	0.864 s	1.76 s	2.05x
cloud	heavy_tailed	8	10,000	none	5	2-2	1.170	1.99x	0.894 s	0.967 s	1.09x
cloud	lattice_ties	8	10,000	none	5	2-2	1.170	1.78x	1.09 s	1.96 s	1.79x
cloud	shell	8	10,000	none	5	4-7	1.183	4.96x	2.63 s	14.8 s	5.57x
cloud	shifted	8	10,000	none	5	7-7	3.086	5.92x	4.81 s	54.3 s	11.13x
dimension	gaussian	2	10,000	none	5	2-4	1.170	0.24x	0.167 s	0.599 s	3.76x
dimension	gaussian	4	10,000	none	5	2-2	1.170	0.69x	0.336 s	1.45 s	4.36x
dimension	gaussian	8	10,000	none	5	2-2	1.170	1.79x	0.864 s	1.77 s	2.05x
dimension	gaussian	16	10,000	none	5	2-2	1.170	2.00x	2.51 s	4.96 s	1.97x
dimension	gaussian	32	10,000	none	5	2-2	1.170	2.00x	9.57 s	21.3 s	2.22x
size	gaussian	8	2,000	none	3	2-2	4.951	1.99x	0.05 s	0.065 s	1.27x
size	gaussian	8	5,000	none	3	2-2	2.160	1.92x	0.25 s	0.369 s	1.48x
size	gaussian	8	10,000	none	3	2-2	1.170	1.79x	0.86 s	1.76 s	2.05x
size	gaussian	8	20,000	none	3	2-2	0.630	1.61x	2.97 s	7.67 s	2.56x
size	gaussian	8	50,000	none	3	2-3	0.288	1.94x	21.7 s	64.3 s	3.02x

Table 8: The certificate by cell. The duality gap is the residual the loop stopped on. Equals the dense solve compares the pairing unit by unit where the dense path could be run, and reads not run where the problem was too large for it.

Sweep	Cloud	Dim	Units	Caliper	Worst duality gap	All certified	Equals the dense solve
caliper	gaussian	8	10,000	boundary	0.00e+00	TRUE	TRUE
caliper	gaussian	8	10,000	mid	0.00e+00	TRUE	TRUE
caliper	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	clustered	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	contested	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	heavy_tailed	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	lattice_ties	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	shell	8	10,000	none	0.00e+00	TRUE	TRUE
cloud	shifted	8	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	2	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	4	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	16	10,000	none	0.00e+00	TRUE	TRUE
dimension	gaussian	32	10,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	2,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	5,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	10,000	none	0.00e+00	TRUE	TRUE
size	gaussian	8	20,000	none	0.00e+00	TRUE	not run
size	gaussian	8	50,000	none	3.64e-12	TRUE	not run

Table 9: Every implicit run in the grid, one row per seed. Seed width is the number of columns the first round gave each row.

Sweep	Cloud	Dim	Units	Caliper	Seed	Rounds	Seed width	Graph %	Seconds	Gap
dimension	gaussian	2	10,000	none	1	2	78	1.170	0.14	0.00e+00
dimension	gaussian	2	10,000	none	2	3	78	1.170	0.175	0.00e+00
dimension	gaussian	2	10,000	none	3	3	78	1.170	0.167	0.00e+00
dimension	gaussian	2	10,000	none	4	2	78	1.170	0.143	0.00e+00
dimension	gaussian	2	10,000	none	5	4	78	1.170	0.183	0.00e+00
dimension	gaussian	4	10,000	none	1	2	78	1.170	0.332	0.00e+00
dimension	gaussian	4	10,000	none	2	2	78	1.170	0.34	0.00e+00
dimension	gaussian	4	10,000	none	3	2	78	1.170	0.339	0.00e+00
dimension	gaussian	4	10,000	none	4	2	78	1.170	0.332	0.00e+00
dimension	gaussian	4	10,000	none	5	2	78	1.170	0.336	0.00e+00
dimension	gaussian	8	10,000	none	1	2	78	1.170	0.865	0.00e+00
dimension	gaussian	8	10,000	none	2	2	78	1.170	0.864	0.00e+00
dimension	gaussian	8	10,000	none	3	2	78	1.170	0.864	0.00e+00
dimension	gaussian	8	10,000	none	4	2	78	1.170	0.869	0.00e+00
dimension	gaussian	8	10,000	none	5	2	78	1.170	0.862	0.00e+00
dimension	gaussian	16	10,000	none	1	2	78	1.170	2.51	0.00e+00
dimension	gaussian	16	10,000	none	2	2	78	1.170	2.51	0.00e+00
dimension	gaussian	16	10,000	none	3	2	78	1.170	2.51	0.00e+00
dimension	gaussian	16	10,000	none	4	2	78	1.170	2.51	0.00e+00

Sweep	Cloud	Dim	Units	Caliper	Seed	Rounds	Seed width	Graph %	Seconds	Gap
dimension	gaussian	16	10,000	none	5	2	78	1.170	2.51	0.00e+00
dimension	gaussian	32	10,000	none	1	2	78	1.170	9.57	0.00e+00
dimension	gaussian	32	10,000	none	2	2	78	1.170	9.57	0.00e+00
dimension	gaussian	32	10,000	none	3	2	78	1.170	9.57	0.00e+00
dimension	gaussian	32	10,000	none	4	2	78	1.170	9.57	0.00e+00
dimension	gaussian	32	10,000	none	5	2	78	1.170	9.57	0.00e+00
cloud	gaussian	8	10,000	none	1	2	78	1.170	0.864	0.00e+00
cloud	gaussian	8	10,000	none	2	2	78	1.170	0.862	0.00e+00
cloud	gaussian	8	10,000	none	3	2	78	1.170	0.868	0.00e+00
cloud	gaussian	8	10,000	none	4	2	78	1.170	0.867	0.00e+00
cloud	gaussian	8	10,000	none	5	2	78	1.170	0.864	0.00e+00
cloud	clustered	8	10,000	none	1	2	78	1.170	0.314	0.00e+00
cloud	clustered	8	10,000	none	2	2	78	1.170	0.324	0.00e+00
cloud	clustered	8	10,000	none	3	2	78	1.170	0.333	0.00e+00
cloud	clustered	8	10,000	none	4	2	78	1.170	0.36	0.00e+00
cloud	clustered	8	10,000	none	5	2	78	1.170	0.313	0.00e+00
cloud	shifted	8	10,000	none	1	7	78	3.062	4.81	0.00e+00
cloud	shifted	8	10,000	none	2	7	78	3.086	4.88	0.00e+00
cloud	shifted	8	10,000	none	3	7	78	3.108	4.92	0.00e+00
cloud	shifted	8	10,000	none	4	7	78	3.099	4.72	0.00e+00
cloud	shifted	8	10,000	none	5	7	78	3.080	4.81	0.00e+00
cloud	lattice_ties	8	10,000	none	1	2	78	1.170	1.1	0.00e+00
cloud	lattice_ties	8	10,000	none	2	2	78	1.170	1.09	0.00e+00
cloud	lattice_ties	8	10,000	none	3	2	78	1.170	1.09	0.00e+00
cloud	lattice_ties	8	10,000	none	4	2	78	1.170	1.09	0.00e+00
cloud	lattice_ties	8	10,000	none	5	2	78	1.170	1.09	0.00e+00
cloud	heavy_tailed	8	10,000	none	1	2	78	1.170	0.894	0.00e+00
cloud	heavy_tailed	8	10,000	none	2	2	78	1.170	0.891	0.00e+00
cloud	heavy_tailed	8	10,000	none	3	2	78	1.170	0.914	0.00e+00
cloud	heavy_tailed	8	10,000	none	4	2	78	1.170	0.894	0.00e+00
cloud	heavy_tailed	8	10,000	none	5	2	78	1.170	0.895	0.00e+00
cloud	contested	8	10,000	none	1	10	78	73.287	134	0.00e+00
cloud	contested	8	10,000	none	2	9	78	73.318	134	0.00e+00
cloud	contested	8	10,000	none	3	10	78	73.347	134	0.00e+00
cloud	contested	8	10,000	none	4	10	78	73.355	135	0.00e+00
cloud	contested	8	10,000	none	5	9	78	73.312	133	0.00e+00
cloud	shell	8	10,000	none	1	4	78	1.172	1.95	0.00e+00
cloud	shell	8	10,000	none	2	7	78	1.200	3.27	0.00e+00
cloud	shell	8	10,000	none	3	6	78	2.825	3.4	0.00e+00
cloud	shell	8	10,000	none	4	5	78	1.183	2.63	0.00e+00
cloud	shell	8	10,000	none	5	5	78	1.176	2.5	0.00e+00
caliper	gaussian	8	10,000	none	1	2	78	1.170	0.865	0.00e+00
caliper	gaussian	8	10,000	none	2	2	78	1.170	0.861	0.00e+00
caliper	gaussian	8	10,000	none	3	2	78	1.170	0.863	0.00e+00
caliper	gaussian	8	10,000	none	4	2	78	1.170	0.866	0.00e+00
caliper	gaussian	8	10,000	none	5	2	78	1.170	0.863	0.00e+00
caliper	gaussian	8	10,000	boundary	1	2	78	1.151	0.91	0.00e+00
caliper	gaussian	8	10,000	boundary	2	2	78	1.145	0.901	0.00e+00
caliper	gaussian	8	10,000	boundary	3	2	78	1.121	0.841	0.00e+00
caliper	gaussian	8	10,000	boundary	4	2	78	1.168	1.05	0.00e+00
caliper	gaussian	8	10,000	boundary	5	2	78	1.165	1	0.00e+00

Sweep	Cloud	Dim	Units	Caliper	Seed	Rounds	Seed width	Graph %	Seconds	Gap
caliper	gaussian	8	10,000	mid	1	2	78	1.166	1.03	0.00e+00
caliper	gaussian	8	10,000	mid	2	2	78	1.125	0.85	0.00e+00
caliper	gaussian	8	10,000	mid	3	2	78	1.142	0.88	0.00e+00
caliper	gaussian	8	10,000	mid	4	2	78	1.167	1.02	0.00e+00
caliper	gaussian	8	10,000	mid	5	2	78	1.163	0.989	0.00e+00
size	gaussian	8	2,000	none	1	2	66	4.951	0.05	0.00e+00
size	gaussian	8	2,000	none	2	2	66	4.951	0.05	0.00e+00
size	gaussian	8	2,000	none	3	2	66	4.951	0.051	0.00e+00
size	gaussian	8	5,000	none	1	2	72	2.160	0.249	0.00e+00
size	gaussian	8	5,000	none	2	2	72	2.160	0.25	0.00e+00
size	gaussian	8	5,000	none	3	2	72	2.160	0.251	0.00e+00
size	gaussian	8	10,000	none	1	2	78	1.170	0.866	0.00e+00
size	gaussian	8	10,000	none	2	2	78	1.170	0.857	0.00e+00
size	gaussian	8	10,000	none	3	2	78	1.170	0.86	0.00e+00
size	gaussian	8	20,000	none	1	2	84	0.630	2.99	0.00e+00
size	gaussian	8	20,000	none	2	2	84	0.630	2.97	0.00e+00
size	gaussian	8	20,000	none	3	2	84	0.630	2.96	0.00e+00
size	gaussian	8	50,000	none	1	2	96	0.288	15.2	0.00e+00
size	gaussian	8	50,000	none	2	3	96	0.288	21.7	0.00e+00
size	gaussian	8	50,000	none	3	3	96	0.288	21.8	3.64e-12

Every cell of the grid finished.

6 Peak memory

Peak resident set size is a property of a process, so each cell here is a fresh R process that loads what it needs, does one matching and exits, measured from outside by the platform's `time` utility. The baseline row of a size is a process that loaded the same packages and matched nothing, and it is what the increment column is measured against. `dense_matrix` builds the distance matrix and stops, which separates the representation from the solver that runs on it.

Table 10: Peak resident set size by arm and problem size. Matrix is the bytes the dense cost matrix itself needs at that shape. Package estimate is what `estimate_dense_matrix_mb()` predicts, which is four times the matrix and is what the memory dispatcher warns from.

Units	Arm	Peak RSS	Over baseline	R heap peak	Matrix	Package estimate	Seconds	Status
2,000	baseline	146 MB	-	80 MB	7 MB	28 MB	0.007	ok
2,000	baseline_alt	153 MB	-	87 MB	7 MB	28 MB	0.008	ok
2,000	dense	265 MB	119 MB	153 MB	7 MB	28 MB	0.167	ok
2,000	dense_matrix	203 MB	56 MB	137 MB	7 MB	28 MB	0.071	ok
2,000	implicit	158 MB	12 MB	102 MB	7 MB	28 MB	0.071	ok

Units	Arm	Peak RSS	Over baseline	R heap peak	Matrix	Package estimate	Seconds	Status
2,000	lazy	151 MB	5 MB	102 MB	7 MB	28 MB	0.068	ok
2,000	MatchIt	530 MB	377 MB	311 MB	7 MB	28 MB	2.38	ok
2,000	optmatch	480 MB	327 MB	318 MB	7 MB	28 MB	1.89	ok
5,000	baseline	146 MB	-	80 MB	44 MB	178 MB	0.008	ok
5,000	baseline__alt	153 MB	-	87 MB	44 MB	178 MB	0.008	ok
5,000	dense	566 MB	420 MB	409 MB	44 MB	178 MB	0.902	ok
5,000	dense__matrix	226 MB	80 MB	161 MB	44 MB	178 MB	0.343	ok
5,000	implicit	175 MB	30 MB	104 MB	44 MB	178 MB	0.313	ok
5,000	lazy	154 MB	8 MB	104 MB	44 MB	178 MB	0.363	ok
5,000	MatchIt	2436 MB	2283 MB	1474 MB	44 MB	178 MB	16.8	ok
5,000	optmatch	2059 MB	1907 MB	1320 MB	44 MB	178 MB	13.4	ok
10,000	baseline	146 MB	-	80 MB	178 MB	711 MB	0.008	ok
10,000	baseline__alt	154 MB	-	87 MB	178 MB	711 MB	0.008	ok
10,000	dense	1429 MB	1283 MB	812 MB	178 MB	711 MB	3.19	ok
10,000	dense__matrix	446 MB	300 MB	374 MB	178 MB	711 MB	1.23	ok
10,000	implicit	205 MB	59 MB	108 MB	178 MB	711 MB	1.11	ok
10,000	lazy	159 MB	14 MB	108 MB	178 MB	711 MB	1.6	ok
10,000	MatchIt	8866 MB	8713 MB	6382 MB	178 MB	711 MB	74.6	ok
10,000	optmatch	7333 MB	7179 MB	4545 MB	178 MB	711 MB	60.9	ok
20,000	baseline	146 MB	-	80 MB	711 MB	2844 MB	0.007	ok
20,000	baseline__alt	153 MB	-	87 MB	711 MB	2844 MB	0.008	ok
20,000	dense	6380 MB	6234 MB	3656 MB	711 MB	2844 MB	10.7	ok
20,000	dense__matrix	241 MB	1095 MB	1145 MB	711 MB	2844 MB	4.3	ok
20,000	implicit	274 MB	128 MB	116 MB	711 MB	2844 MB	3.92	ok
20,000	lazy	168 MB	22 MB	116 MB	711 MB	2844 MB	6.42	ok
20,000	MatchIt	8794 MB	8640 MB	7121 MB	711 MB	2844 MB	114	error: (from optmatch) result would exceed 2^31-1 bytes

Units	Arm	Peak RSS	Over baseline	R heap peak	Matrix	Package estimate	Seconds	Status
20,000	optmatch	27581 MB	27428 MB	19331 MB	711 MB	2844 MB	281	ok

Table 11: The four-fold estimate against the measurement. The estimate is deliberately conservative: it exists to avoid under-warning before a large allocation, not to predict the peak.

Units	Matrix	Measured over baseline	Measured / matrix	Package estimate
2,000	7 MB		7.96	28 MB
5,000	44 MB		1.81	178 MB
10,000	178 MB		1.69	711 MB
20,000	711 MB		1.54	2844 MB

7 Representation against solver

The implicit mode is a representation and a solver at once: its restricted master is the flow model, and naming a solver under it is an error rather than a setting. Reading its timing as a representation comparison would therefore attribute the flow solver’s behaviour to the representation. The table below is the decomposition: down a column, one solver over two representations; across a row, one representation under two solvers.

Table 12: Wall-clock seconds by representation and solver. Dense and lazy differ only in whether the matrix is materialised; dense and dense-flow differ only in the solver; implicit is the mode as a user meets it.

Units	Dense, JV	Lazy, JV	Dense, flow	Implicit
500	22 ms	8 ms	21 ms	12 ms
2,000	161 ms	53 ms	196 ms	56 ms
5,000	837 ms	351 ms	1.14 s	299 ms
10,000	3.14 s	1.59 s	4.17 s	1.09 s
20,000	not run	6.38 s	not run	3.9 s
50,000	not run	62 s	not run	20.2 s

8 Scaling, every run

The scaling table in the article reports the median across instances. These are the runs behind it, one row per repetition.

Table 13: Every scaling run. Instance is an independently generated problem of that size; rep is a repetition on that problem. A run that failed or timed out carries its status and no time.

Units	Package	Instance	Seed	Rep	Seconds	Status
500	couplr	1	20261015	1	0.018	ok
500	couplr	1	20261015	2	0.017	ok
500	couplr	1	20261015	3	0.017	ok

Units	Package	Instance	Seed	Rep	Seconds	Status
500	couplr	2	21261018	1	0.017	ok
500	couplr	2	21261018	2	0.018	ok
500	couplr	2	21261018	3	0.017	ok
500	couplr	3	22261021	1	0.017	ok
500	couplr	3	22261021	2	0.016	ok
500	couplr	3	22261021	3	0.016	ok
500	couplr	4	23261024	1	0.019	ok
500	couplr	4	23261024	2	0.019	ok
500	couplr	4	23261024	3	0.019	ok
500	couplr	5	24261027	1	0.021	ok
500	couplr	5	24261027	2	0.021	ok
500	couplr	5	24261027	3	0.021	ok
500	MatchIt	1	20261015	1	0.163	ok
500	MatchIt	1	20261015	2	0.166	ok
500	MatchIt	1	20261015	3	0.166	ok
500	MatchIt	2	21261018	1	0.13	ok
500	MatchIt	2	21261018	2	0.131	ok
500	MatchIt	2	21261018	3	0.131	ok
500	MatchIt	3	22261021	1	0.132	ok
500	MatchIt	3	22261021	2	0.134	ok
500	MatchIt	3	22261021	3	0.13	ok
500	MatchIt	4	23261024	1	0.137	ok
500	MatchIt	4	23261024	2	0.14	ok
500	MatchIt	4	23261024	3	0.139	ok
500	MatchIt	5	24261027	1	0.134	ok
500	MatchIt	5	24261027	2	0.133	ok
500	MatchIt	5	24261027	3	0.134	ok
500	optmatch	1	20261015	1	0.107	ok
500	optmatch	1	20261015	2	0.108	ok
500	optmatch	1	20261015	3	0.108	ok
500	optmatch	2	21261018	1	0.111	ok
500	optmatch	2	21261018	2	0.112	ok
500	optmatch	2	21261018	3	0.111	ok
500	optmatch	3	22261021	1	0.12	ok
500	optmatch	3	22261021	2	0.12	ok
500	optmatch	3	22261021	3	0.128	ok
500	optmatch	4	23261024	1	0.119	ok
500	optmatch	4	23261024	2	0.121	ok
500	optmatch	4	23261024	3	0.122	ok
500	optmatch	5	24261027	1	0.153	ok
500	optmatch	5	24261027	2	0.163	ok
500	optmatch	5	24261027	3	0.152	ok
2,000	couplr	1	20262515	1	0.163	ok
2,000	couplr	1	20262515	2	0.164	ok
2,000	couplr	1	20262515	3	0.163	ok
2,000	couplr	2	21262518	1	0.136	ok
2,000	couplr	2	21262518	2	0.138	ok
2,000	couplr	2	21262518	3	0.137	ok
2,000	couplr	3	22262521	1	0.138	ok
2,000	couplr	3	22262521	2	0.14	ok
2,000	couplr	3	22262521	3	0.139	ok
2,000	couplr	4	23262524	1	0.143	ok

Units	Package	Instance	Seed	Rep	Seconds	Status
2,000	couplr	4	23262524	2	0.143	ok
2,000	couplr	4	23262524	3	0.143	ok
2,000	couplr	5	24262527	1	0.14	ok
2,000	couplr	5	24262527	2	0.141	ok
2,000	couplr	5	24262527	3	0.139	ok
2,000	MatchIt	1	20262515	1	2.12	ok
2,000	MatchIt	1	20262515	2	2.164	ok
2,000	MatchIt	1	20262515	3	2.156	ok
2,000	MatchIt	2	21262518	1	2.037	ok
2,000	MatchIt	2	21262518	2	2.035	ok
2,000	MatchIt	2	21262518	3	2.055	ok
2,000	MatchIt	3	22262521	1	2.209	ok
2,000	MatchIt	3	22262521	2	2.287	ok
2,000	MatchIt	3	22262521	3	2.283	ok
2,000	MatchIt	4	23262524	1	2.137	ok
2,000	MatchIt	4	23262524	2	2.132	ok
2,000	MatchIt	4	23262524	3	2.185	ok
2,000	MatchIt	5	24262527	1	2.096	ok
2,000	MatchIt	5	24262527	2	2.13	ok
2,000	MatchIt	5	24262527	3	2.117	ok
2,000	optmatch	1	20262515	1	1.769	ok
2,000	optmatch	1	20262515	2	1.765	ok
2,000	optmatch	1	20262515	3	1.795	ok
2,000	optmatch	2	21262518	1	1.617	ok
2,000	optmatch	2	21262518	2	1.613	ok
2,000	optmatch	2	21262518	3	1.619	ok
2,000	optmatch	3	22262521	1	1.779	ok
2,000	optmatch	3	22262521	2	1.794	ok
2,000	optmatch	3	22262521	3	1.804	ok
2,000	optmatch	4	23262524	1	1.749	ok
2,000	optmatch	4	23262524	2	1.787	ok
2,000	optmatch	4	23262524	3	1.751	ok
2,000	optmatch	5	24262527	1	1.614	ok
2,000	optmatch	5	24262527	2	1.672	ok
2,000	optmatch	5	24262527	3	1.741	ok
5,000	couplr	1	20265515	1	0.751	ok
5,000	couplr	1	20265515	2	0.753	ok
5,000	couplr	2	21265518	1	0.823	ok
5,000	couplr	2	21265518	2	0.819	ok
5,000	couplr	3	22265521	1	0.832	ok
5,000	couplr	3	22265521	2	0.836	ok
5,000	couplr	4	23265524	1	0.776	ok
5,000	couplr	4	23265524	2	0.782	ok
5,000	couplr	5	24265527	1	0.786	ok
5,000	couplr	5	24265527	2	0.792	ok
5,000	MatchIt	1	20265515	1	16.56	ok
5,000	MatchIt	1	20265515	2	16.78	ok
5,000	MatchIt	2	21265518	1	16.35	ok
5,000	MatchIt	2	21265518	2	16.16	ok
5,000	MatchIt	3	22265521	1	16.25	ok
5,000	MatchIt	3	22265521	2	16.35	ok
5,000	MatchIt	4	23265524	1	16.22	ok

Units	Package	Instance	Seed	Rep	Seconds	Status
5,000	MatchIt	4	23265524	2	16.2	ok
5,000	MatchIt	5	24265527	1	15.98	ok
5,000	MatchIt	5	24265527	2	15.8	ok
5,000	optmatch	1	20265515	1	13.19	ok
5,000	optmatch	1	20265515	2	13.18	ok
5,000	optmatch	2	21265518	1	12.75	ok
5,000	optmatch	2	21265518	2	12.78	ok
5,000	optmatch	3	22265521	1	12.76	ok
5,000	optmatch	3	22265521	2	12.79	ok
5,000	optmatch	4	23265524	1	12.73	ok
5,000	optmatch	4	23265524	2	12.73	ok
5,000	optmatch	5	24265527	1	12.43	ok
5,000	optmatch	5	24265527	2	12.48	ok
10,000	couplr	1	20270515	1	3.153	ok
10,000	couplr	1	20270515	2	3.163	ok
10,000	couplr	2	21270518	1	3.031	ok
10,000	couplr	2	21270518	2	3.042	ok
10,000	couplr	3	22270521	1	3.284	ok
10,000	couplr	3	22270521	2	3.285	ok
10,000	MatchIt	1	20270515	1	74.17	ok
10,000	MatchIt	1	20270515	2	74.17	ok
10,000	MatchIt	2	21270518	1	74.47	ok
10,000	MatchIt	2	21270518	2	74.56	ok
10,000	MatchIt	3	22270521	1	72.79	ok
10,000	MatchIt	3	22270521	2	72.77	ok
10,000	optmatch	1	20270515	1	60.41	ok
10,000	optmatch	1	20270515	2	60.51	ok
10,000	optmatch	2	21270518	1	60.59	ok
10,000	optmatch	2	21270518	2	60.54	ok
10,000	optmatch	3	22270521	1	59.19	ok
10,000	optmatch	3	22270521	2	59.17	ok
20,000	couplr	1	20280515	1	10.7	ok
20,000	couplr	2	21280518	1	12.54	ok
20,000	MatchIt	1	20280515	1	-	error: (from optmatch) result would exceed 2 ³¹ -1 bytes
20,000	MatchIt	2	21280518	1	-	error: (from optmatch) result would exceed 2 ³¹ -1 bytes
20,000	optmatch	1	20280515	1	280.6	ok
20,000	optmatch	2	21280518	1	292.8	ok
50,000	couplr	1	20310515	1	76.25	ok
50,000	MatchIt	1	20310515	1	-	timeout
50,000	optmatch	1	20310515	1	-	timeout

9 Software design

The article describes the three layers the package is organised in, the dispatch rules, the memory modes and how the package is verified. This section carries the object model, the C++ organisation and the dependency footprint.

9.1 Object model

The object model is S3 throughout. Matching results carry the class vector `c("matching_result", "couplr_result")`, with the design-specific classes (`full_matching_result`, `cem_result`, `subclass_result`) sharing the `couplr_result` parent, which is what lets `balance_diagnostics()`, `join_matched()`, `match_data()` and the `bal.tab()` methods dispatch on it once rather than being rewritten per design. Results are plain lists of tibbles, so a user who wants something the package does not provide can reach in and take it. `lap_solve()` returns a tibble subclassed as `lap_solve_result` with `source`, `target` and `cost` columns, so an optimum is usable in a plot or a join without an extraction step, while `get_total_cost()` and `get_method_used()` recover the metadata carried in attributes.

9.2 C++ organisation

All nineteen solvers are written from scratch in C++ and exposed through Rcpp, with `src/core` holding the cost-source types and shared utilities with no Rcpp dependency, `src/solvers` one algorithm per file with a thin `_rcpp.cpp` wrapper each, `src/flow` the flow model, its two solvers and the edge-generation loop, and `src/interface` the conversion between R objects and the internal types. The separation means the solver bodies are ordinary C++ that can be compiled and tested outside R, which is what the C++ test suite does.

Solvers are templated on their cost source rather than written against a concrete matrix. A cost source provides `at(i, j)`, `allowed(i, j)` and `empty()`, and two types satisfy that interface: `CostMatrix`, which owns a dense flat array plus a mask, and `LazyCostMatrix`, which owns the two feature matrices and computes C_{ij} on demand from the requested metric, applying calipers and the distance ceiling in `allowed()`. A handful of hot loops index the dense mask array directly for speed, which the lazy type cannot support; these are guarded by a `supports_raw_mask` trait and an `if constexpr` branch rather than by duplicating the solver. One solver body therefore serves both cost representations, and the pricing oracle of the edge-generation loop needs `at(i, j)` and `allowed(i, j)` and nothing else, so it consumes the same concept and adds no third representation of the costs.

`LazyCostMatrix` also bakes in the transformations that the dense path applies in a separate preparation pass, namely mapping forbidden entries to a large sentinel and negating costs when maximising. Doing this at construction is what allows the untouched solver body to work on either type.

9.3 Dependency footprint

`couplr` declares Rcpp, tibble, dplyr, rlang, purrr and `methods` in `Imports`, and Rcpp alone in `LinkingTo`. The recursive closure of hard dependencies is 17 non-base packages, so a default installation brings 18 packages including `couplr` itself. The comparable figures on the same day were 18 for `optmatch`, 8 for `MatchIt` and 7 for `designmatch`.

Two choices sit behind that number. First, we use no external solver: all nineteen algorithms are package source, so there is no dependency on an LP or MIP library and no system solver to install. That choice also keeps `LinkingTo` to Rcpp alone, which matters because every entry there forces a compile of that package before `couplr` can be built from source; `couplr` declares only what its headers actually include. Second, everything that serves a single feature lives in `Suggests` and is loaded at call time. The animation and image-morphing functions, the plotting helpers, the interoperability layer for `MatchIt`, `optmatch` and `cobalt`, and the parallel back end are all optional in this sense, so a user who only wants to match pays for none of them.